

TRADITIONAL OR AGILE PROJECT MANAGEMENT, WHICH ONE IS BETTER FOR YOUR ORGANIZATION?

- This approach is based on The PMBOK Guide
- Supporting by other resources



SESSION 1



PROJECT MANAGEMENT OVERVIEW

A PMBOK GUIDE APPROACH

What is a Project ?

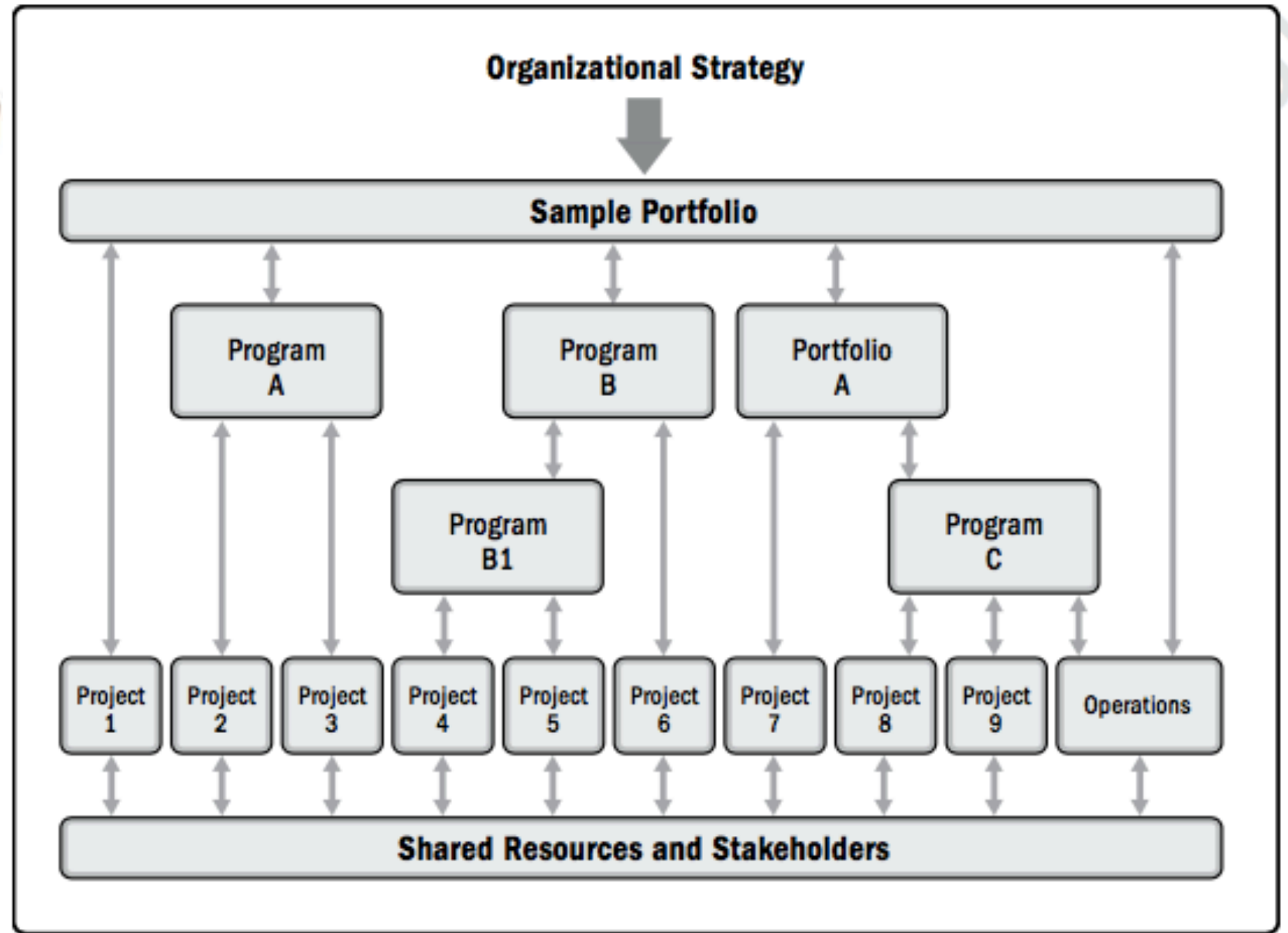
A project is a temporary endeavor undertaken to create a unique product, service, or result.

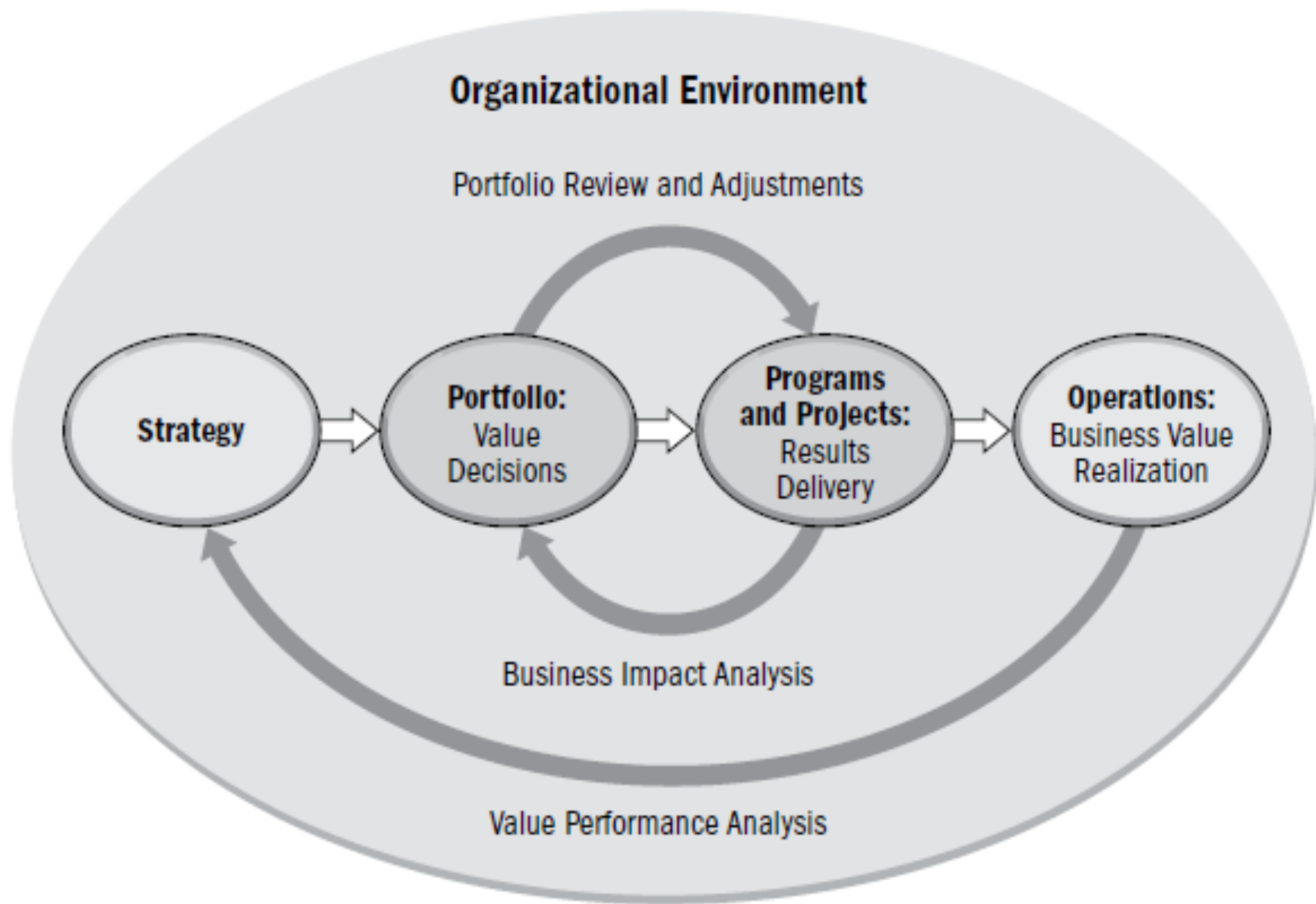
A program is defined as a group of related projects, subsidiary programs, and program activities managed in a coordinated manner to obtain benefits not available from managing them individually.

A portfolio is defined as projects, programs, subsidiary portfolios, and operations managed as a group to achieve strategic objectives.



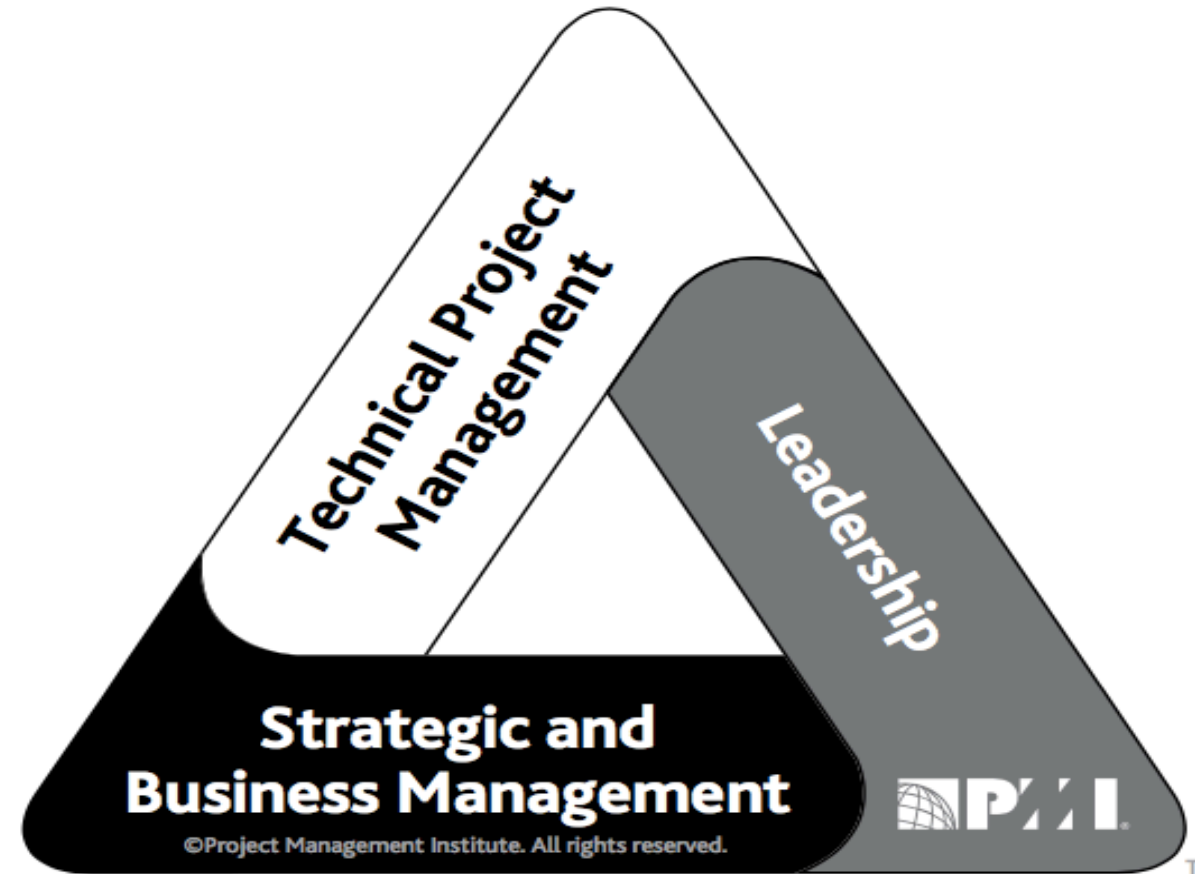
Revenue



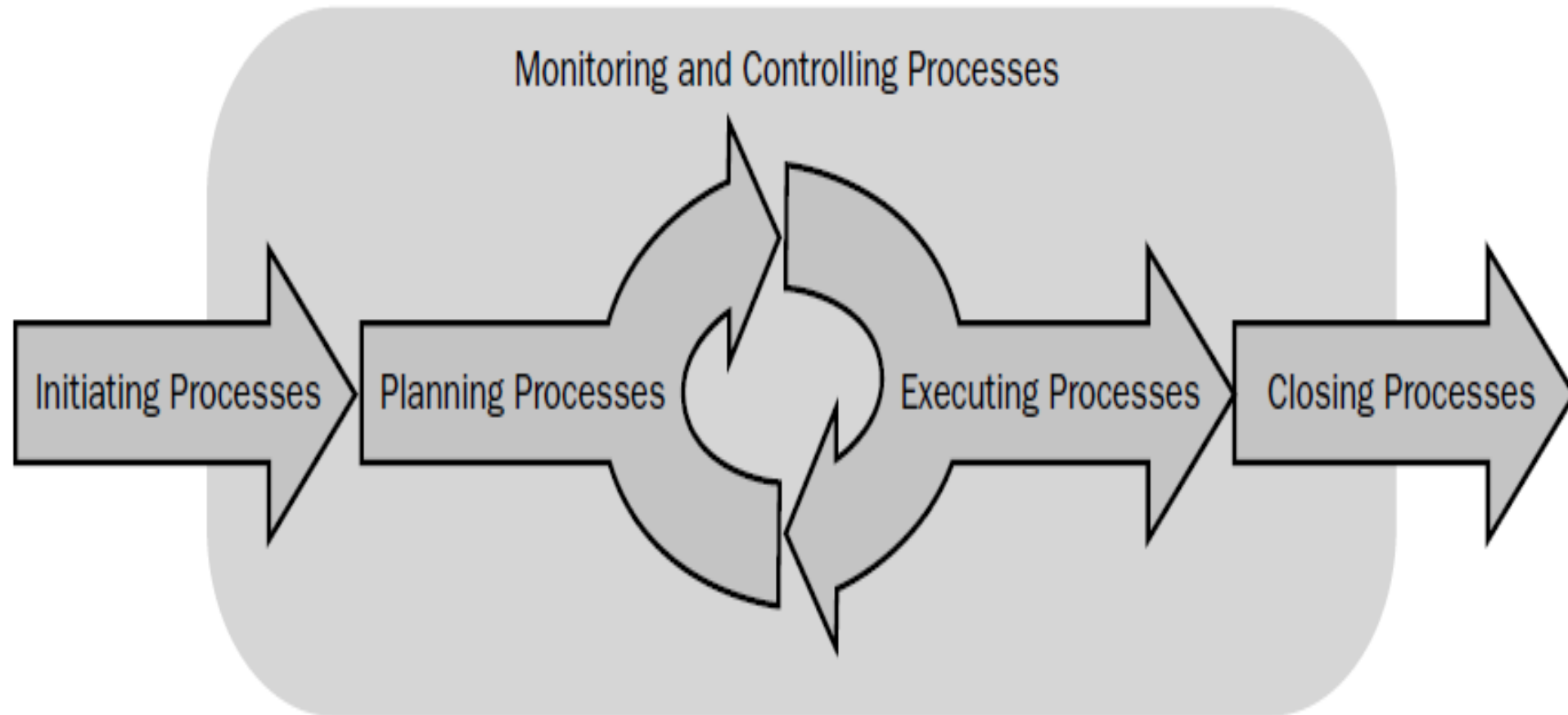


Project Manager Competence

The PMI Talent Triangle®



Project Management Process



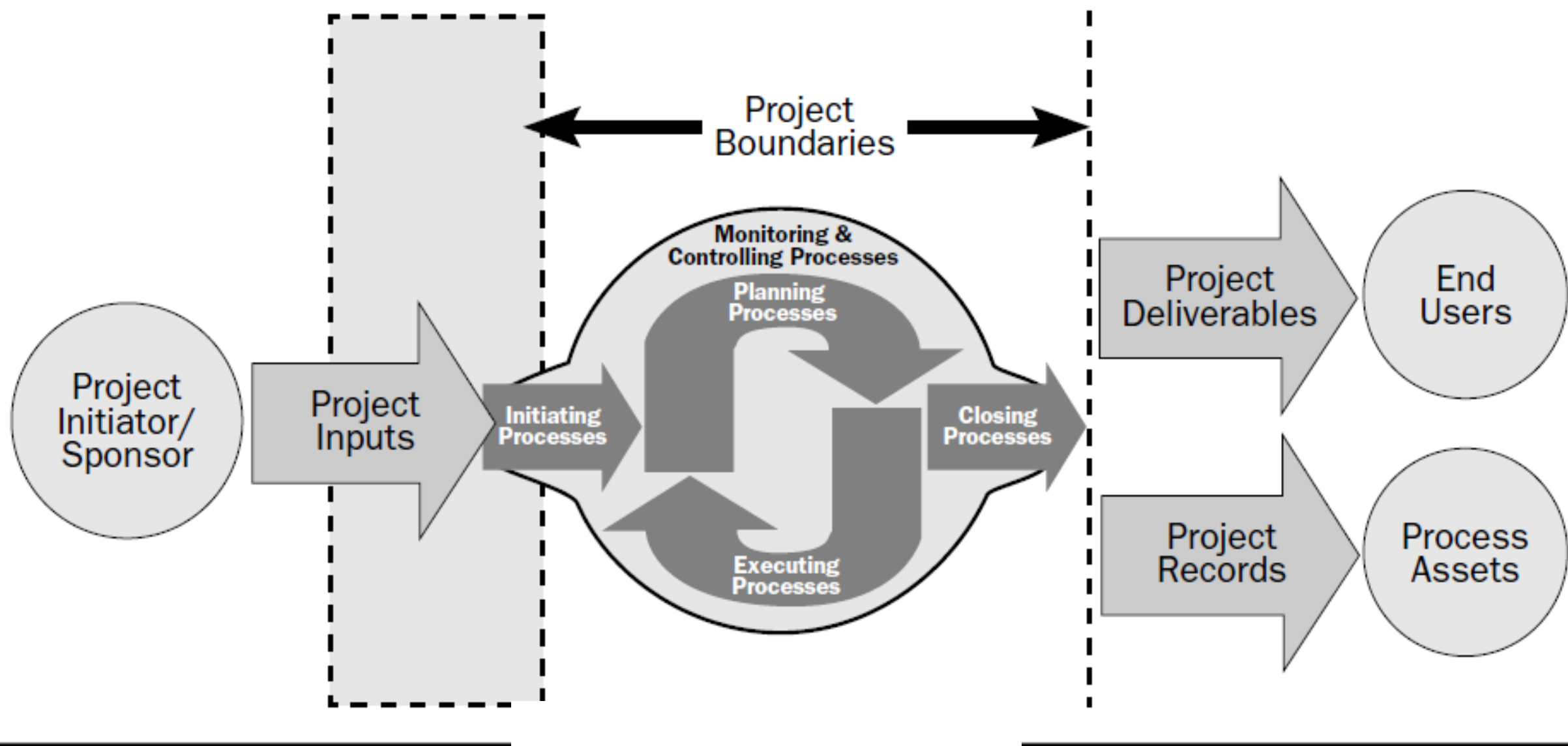
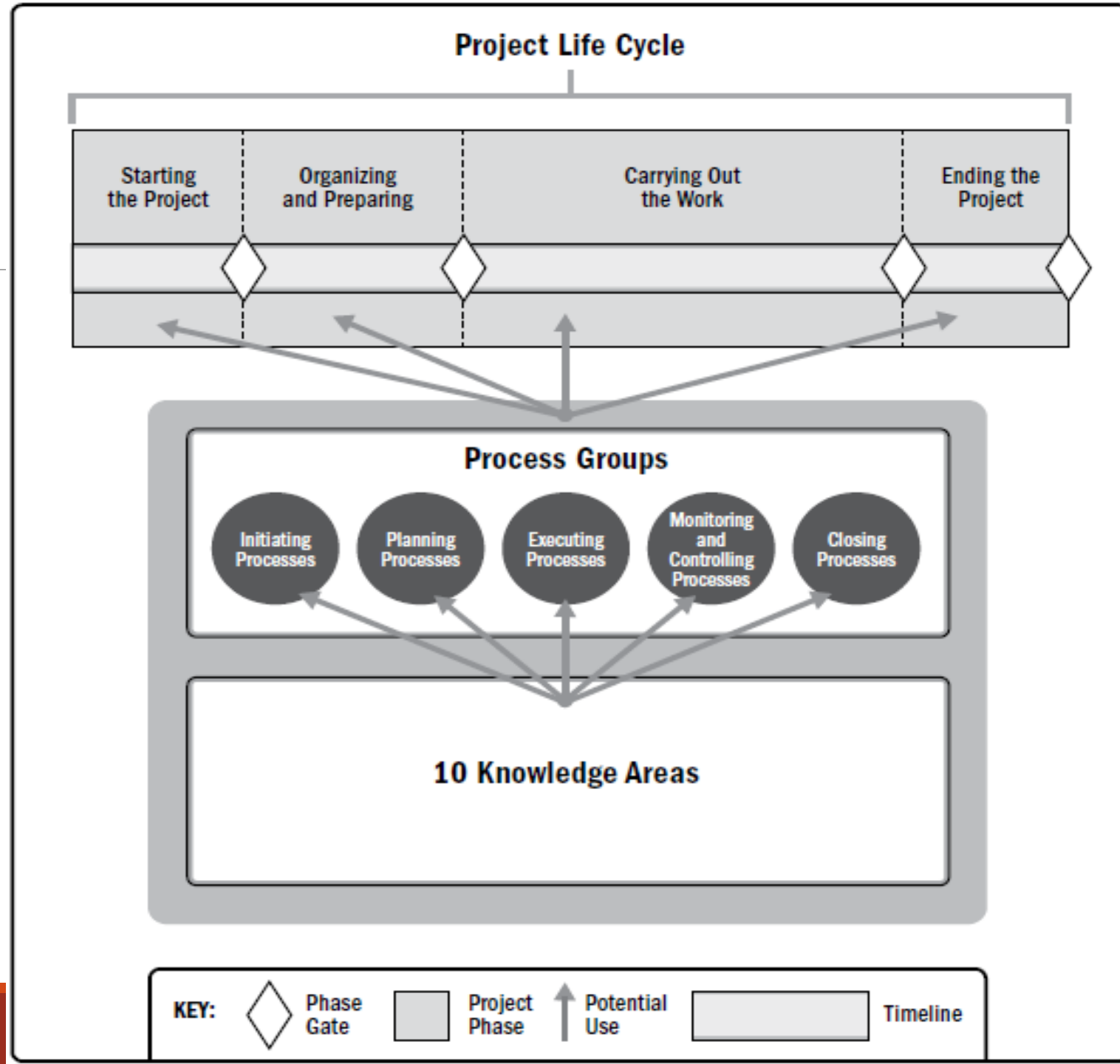


Figure 3-4. Project Boundaries

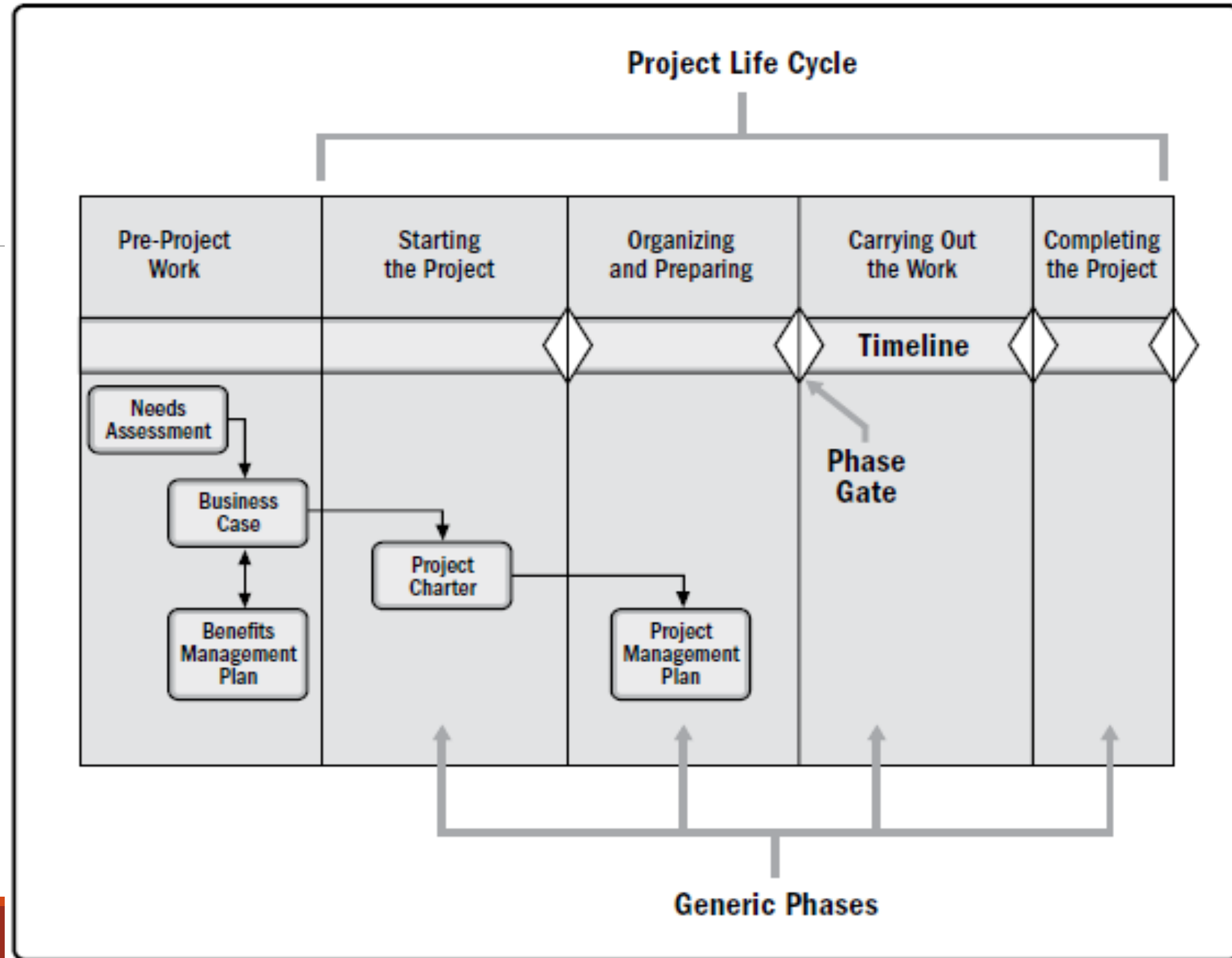
Description of *PMBOK® Guide* Key Components

<i>PMBOK® Guide</i> Key Component	Brief Description
Project life cycle (Section 1.2.4.1)	The series of phases that a project passes through from its start to its completion.
Project phase (Section 1.2.4.2)	A collection of logically related project activities that culminates in the completion of one or more deliverables.
Phase gate (Section 1.2.4.3)	A review at the end of a phase in which a decision is made to continue to the next phase, to continue with modification, or to end a program or project.
Project management processes (Section 1.2.4.4)	A systematic series of activities directed toward causing an end result where one or more inputs will be acted upon to create one or more outputs.
Project Management Process Group (Section 1.2.4.5)	A logical grouping of project management inputs, tools and techniques, and outputs. The Project Management Process Groups include Initiating, Planning, Executing, Monitoring and Controlling, and Closing. Project Management Process Groups are not project phases.
Project Management Knowledge Area (Section 1.2.4.6)	An identified area of project management defined by its knowledge requirements and described in terms of its component processes, practices, inputs, outputs, tools, and techniques.

Interrelationship of *PMBOK® Guide* Key Components in Projects



Interrelationship of
Needs Assessment
and Critical
Business/Project
Documents



Knowledge Areas	Project Management Process Groups				
	Initiating Process Group	Planning Process Group	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	

9. Project Resource Management		9.1 Plan Resource Management 9.2 Estimate Activity Resources	9.3 Acquire Resources 9.4 Develop Team 9.5 Manage Team	9.6 Control Resources	
10. Project Communications Management		10.1 Plan Communications Management	10.2 Manage Communications	10.3 Monitor Communications	
11. Project Risk Management		11.1 Plan Risk Management 11.2 Identify Risks 11.3 Perform Qualitative Risk Analysis 11.4 Perform Quantitative Risk Analysis 11.5 Plan Risk Responses	11.6 Implement Risk Responses	11.7 Monitor Risks	
12. Project Procurement Management		12.1 Plan Procurement Management	12.2 Conduct Procurements	12.3 Control Procurements	
13. Project Stakeholder Management	13.1 Identify Stakeholders	13.2 Plan Stakeholder Engagement	13.3 Manage Stakeholder Engagement	13.4 Monitor Stakeholder Engagement	



SESSION 2



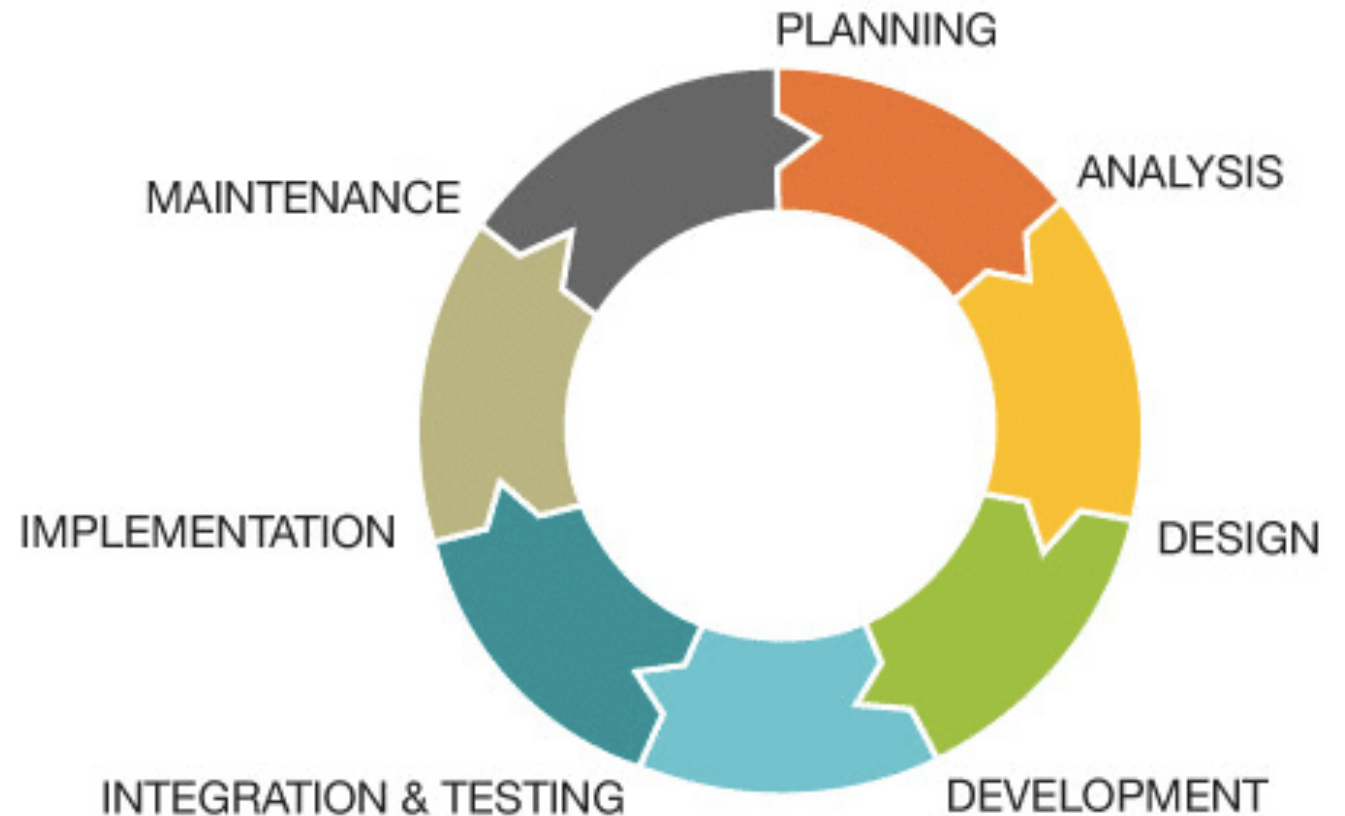
IT PROJECT AND CHALLENGES

System Development Life Cycle

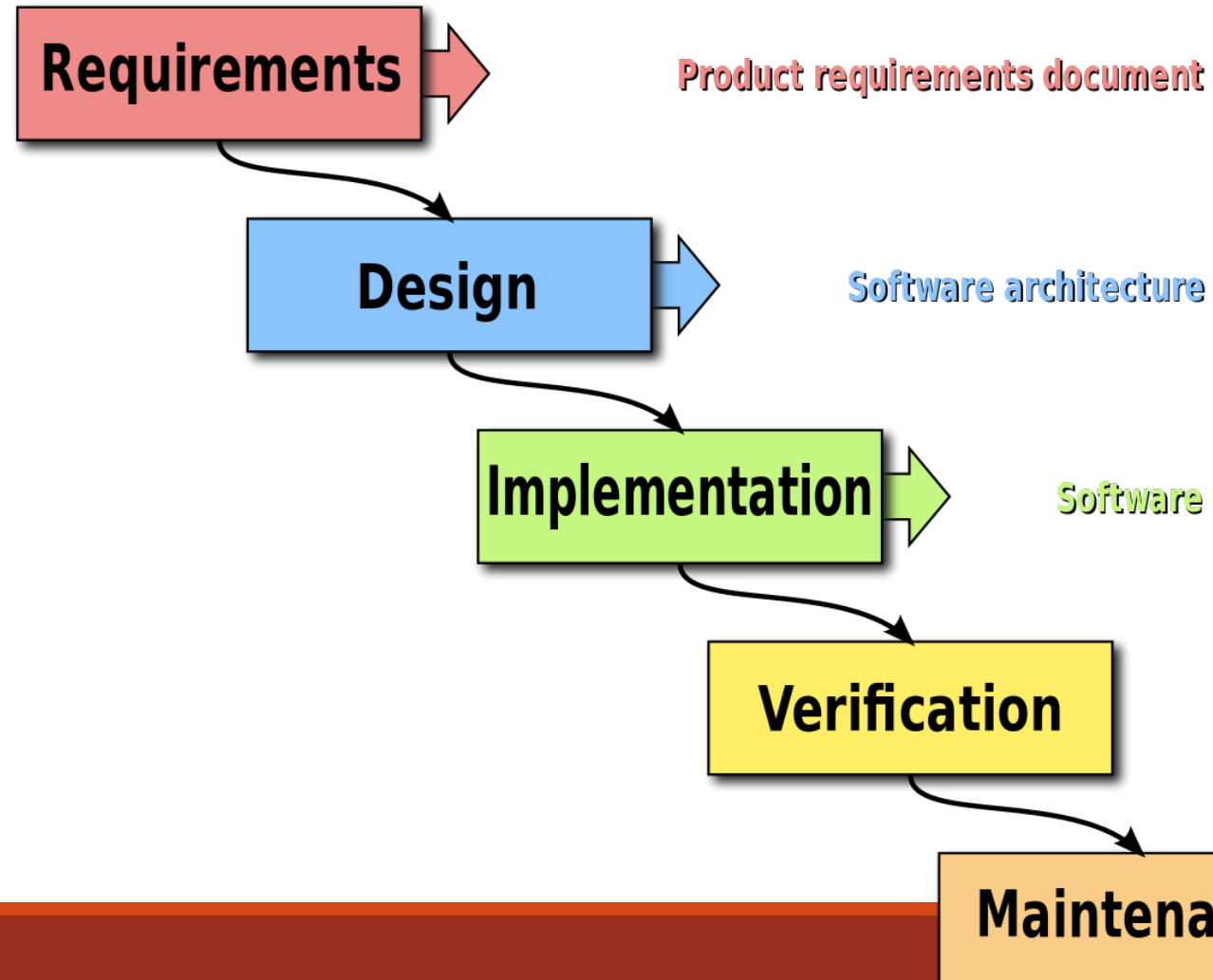
Defines all of the sub phases and deliverables associated with the Execute and Control Project Management Life Cycle phase.

Number of Ways to implement the SDLC

- **Waterfall**
- **Agile**



Waterfall Model



Rational Unified Process

DISCIPLINES

BUSINESS MODELLING

REQUIREMENTS

ANALYSIS & DESIGN

IMPLEMENTATION

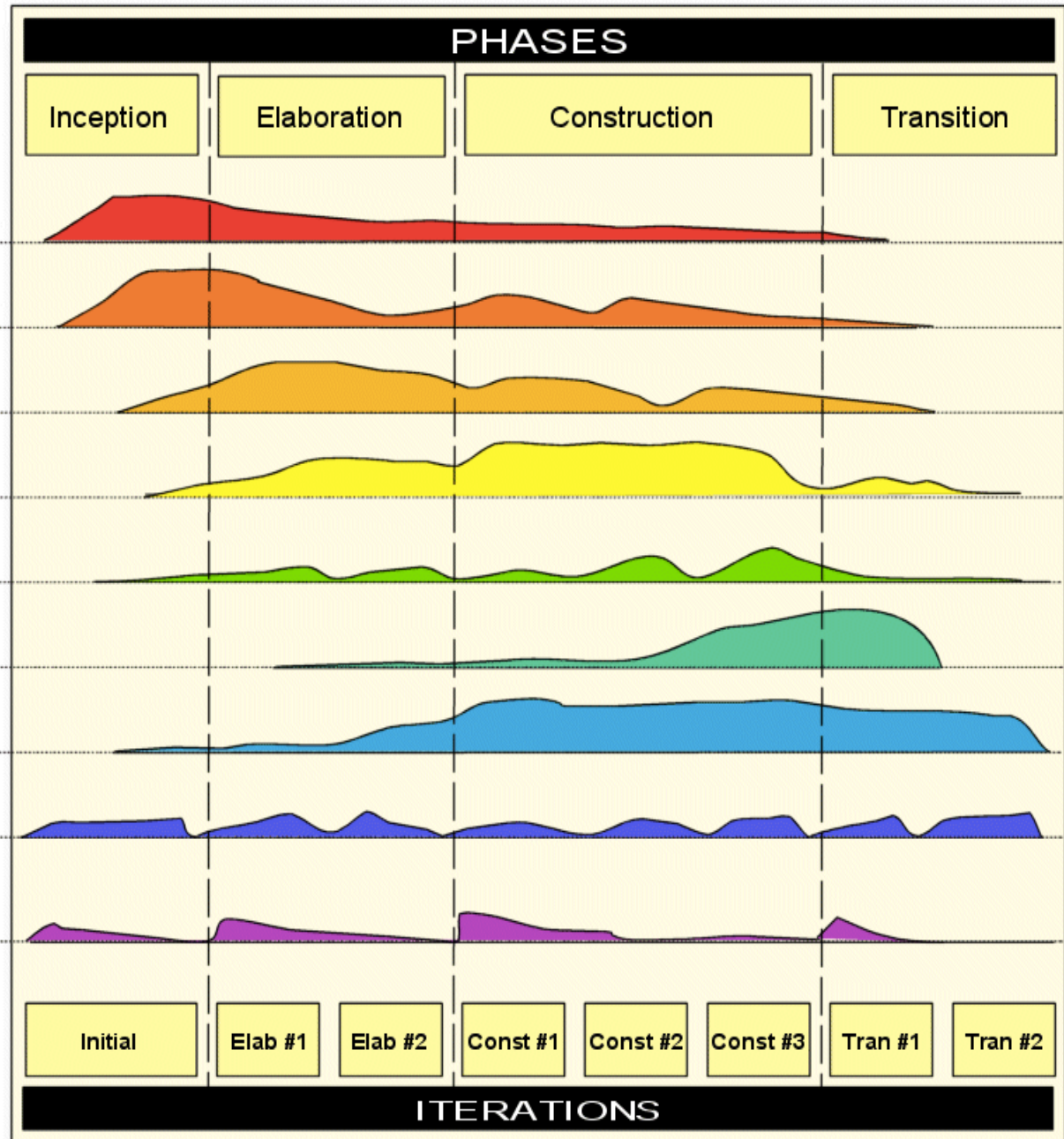
TEST

DEPLOYMENT

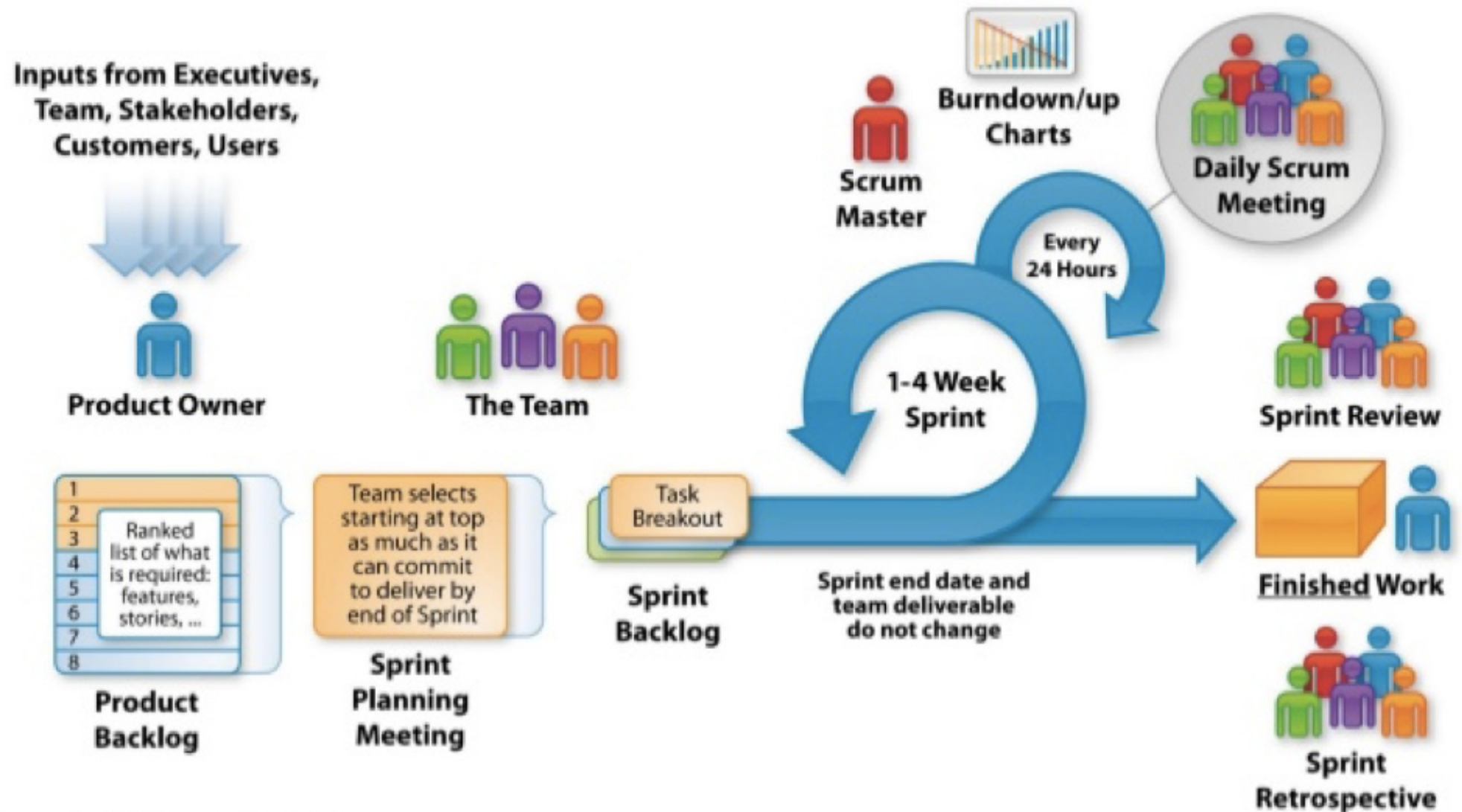
CONFIGURATION & CHANGE MANAGEMENT

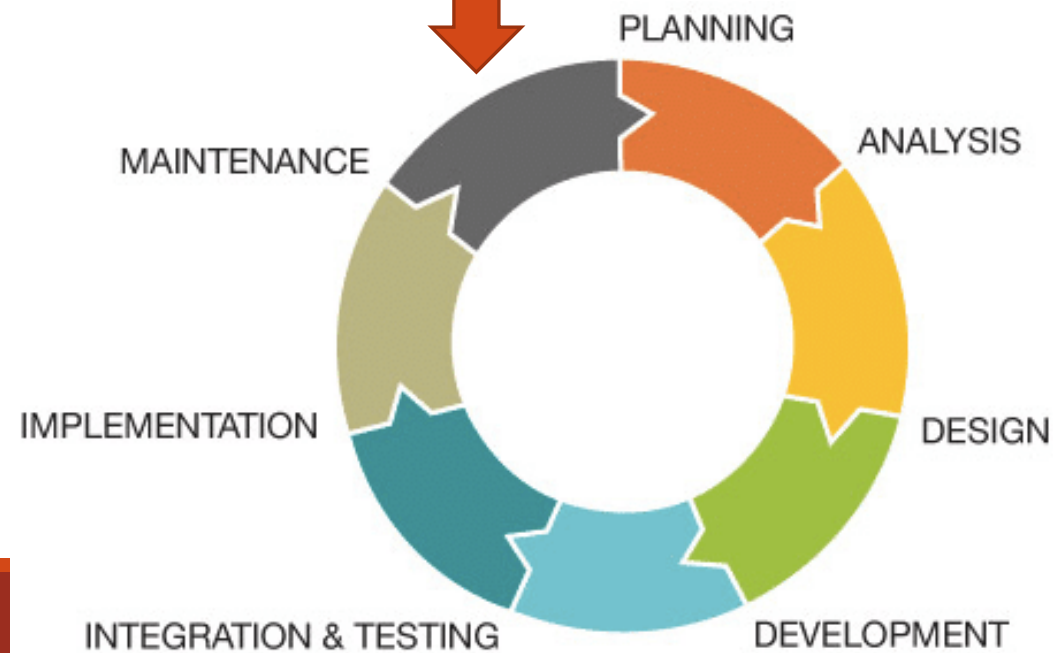
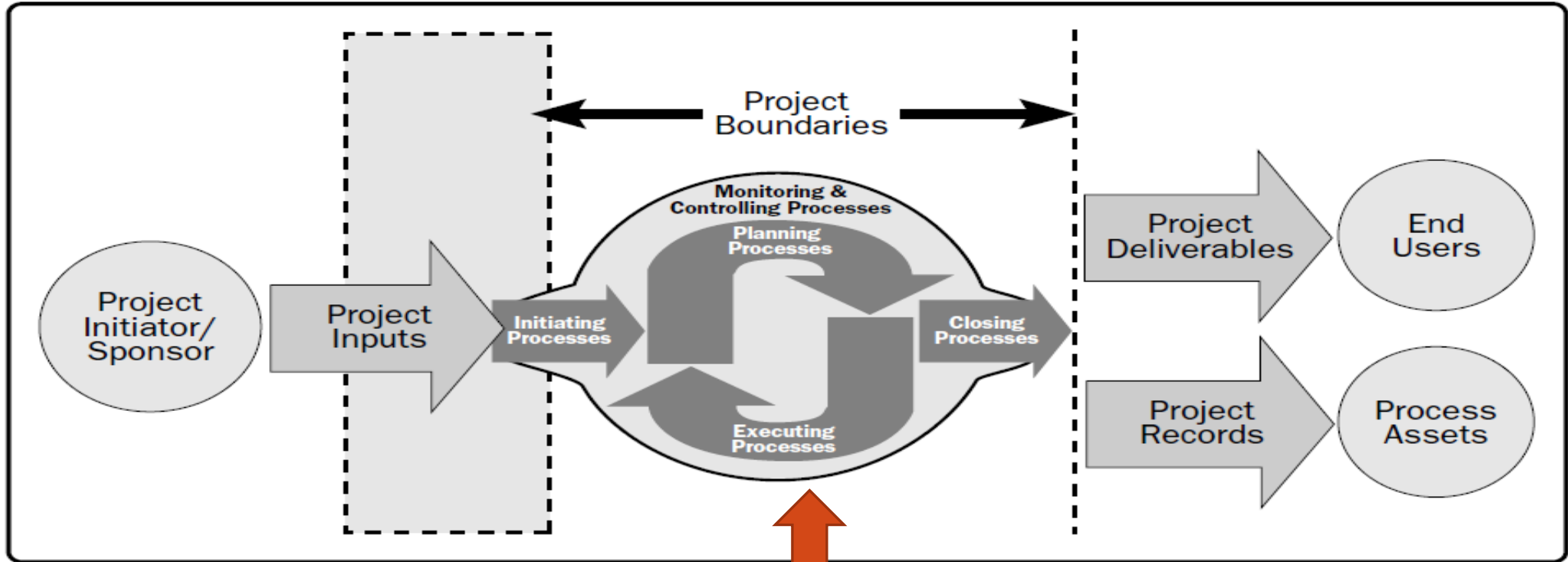
PROJECT MANAGEMENT

ENVIRONMENT



Agile: Scrum Framework



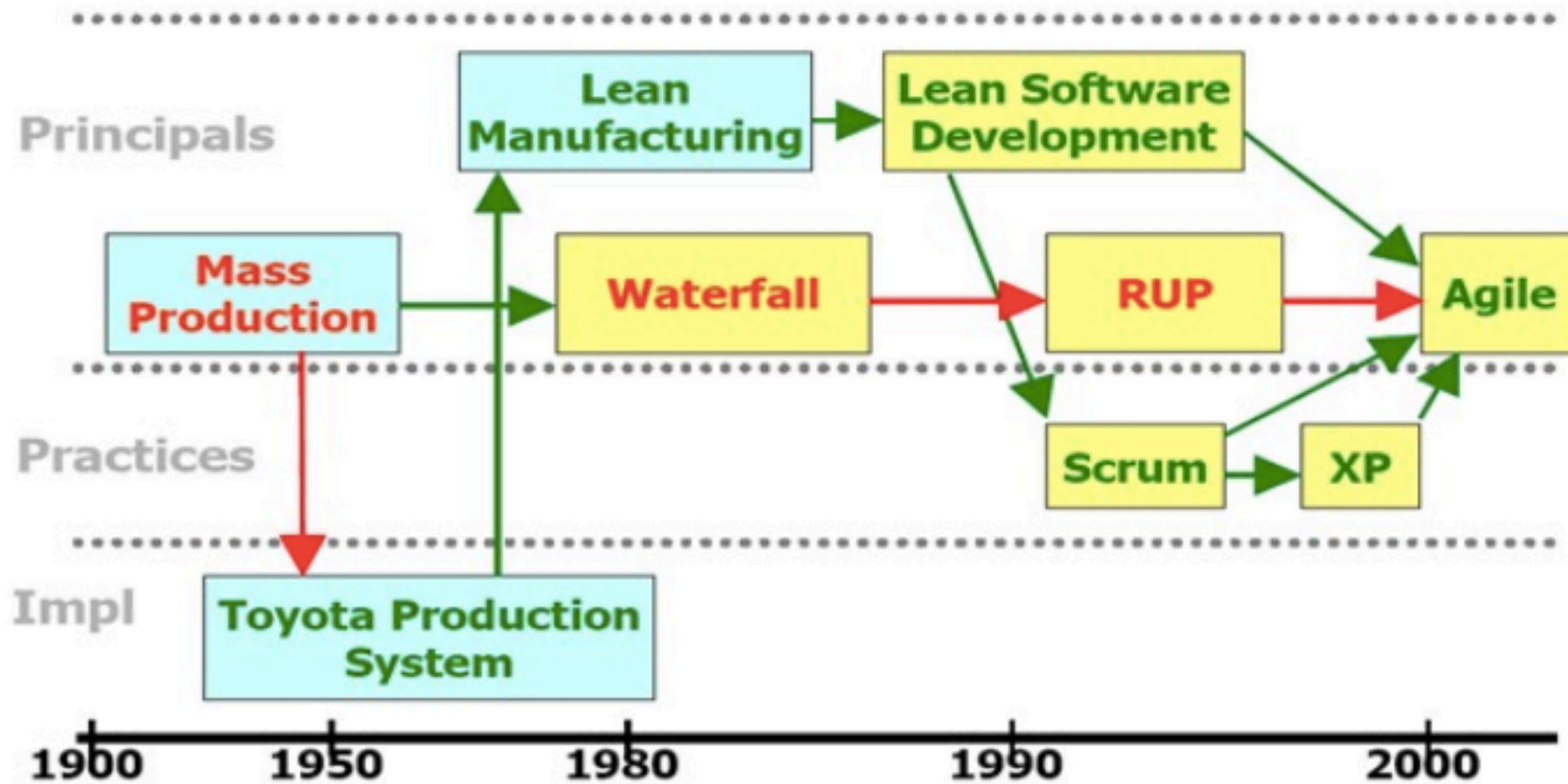


SESSION 3



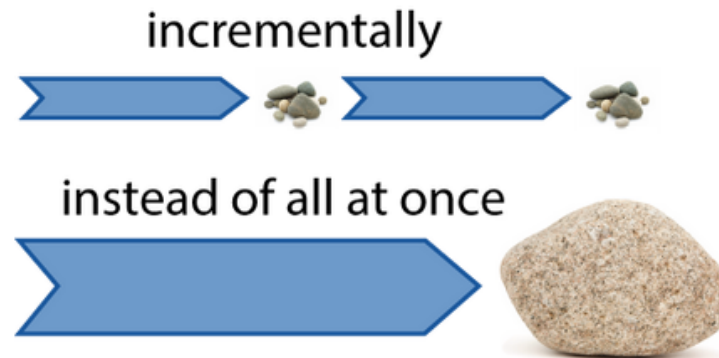
AGILE APPROACH

History of Agile

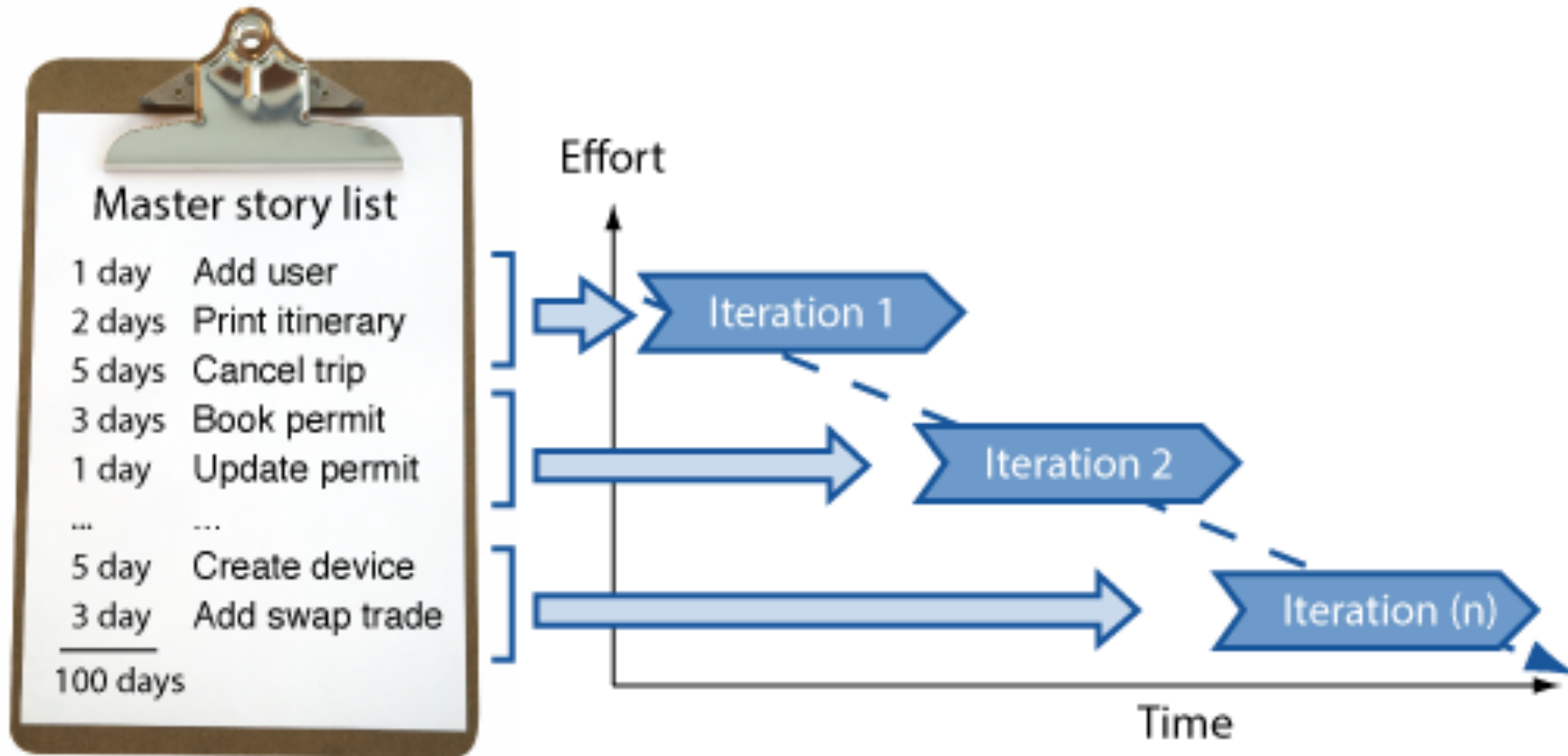


What is Agile?

Agile is a **time boxed, iterative approach** to software delivery that builds software incrementally from the start of the project, instead of trying to deliver it all at once near the end.



What is Agile?



History of Agile

The important highlights as I see them...

1980 – Toyota Production System (TPS) becomes popular in manufacturing, the beginnings of 'Lean'

1990 – Wicked Problems, Righteous Solutions by Peter DeGrace & Leslie Hulet Stahl discusses the scrum approach and other Lean methods for Software Development

1995 – Adaptive Software Development, Feature Driven Development, and Dynamic Systems Development Method (DSDM), all begin to evolve

1999 – Extreme Programming Explained is published by Kent Beck

2001 – Ken Schwaber & Mike Beedle publish the now famous: Agile Software Development with Scrum

2005ish – Agile Unified Process (AUP) - Scott Ambler, Open Unified Process (OpenUp) - eclipse project, Essential Unified Process (EssUP) - Ivar Jacobson; all start to emerge

1980s

1990s

2000s

1986 – The New New Product Development Game by by Hirotaka Takeuchi & Ikujiro Nonaka is published in Harvard Business Review, mentioning Rugby/Scrum approach to manufacturing

1996 – Extreme Programming (XP) begins on the Chrysler Payroll Project

1995 – Ken Schwaber & Jeff Sutherland present Scrum methodology for Business object design and implementation at OOPSLA '95 workshop proceedings

2006 – Implementing Lean Software Development, by Mary and Tom Poppendieck, introduces Kanban

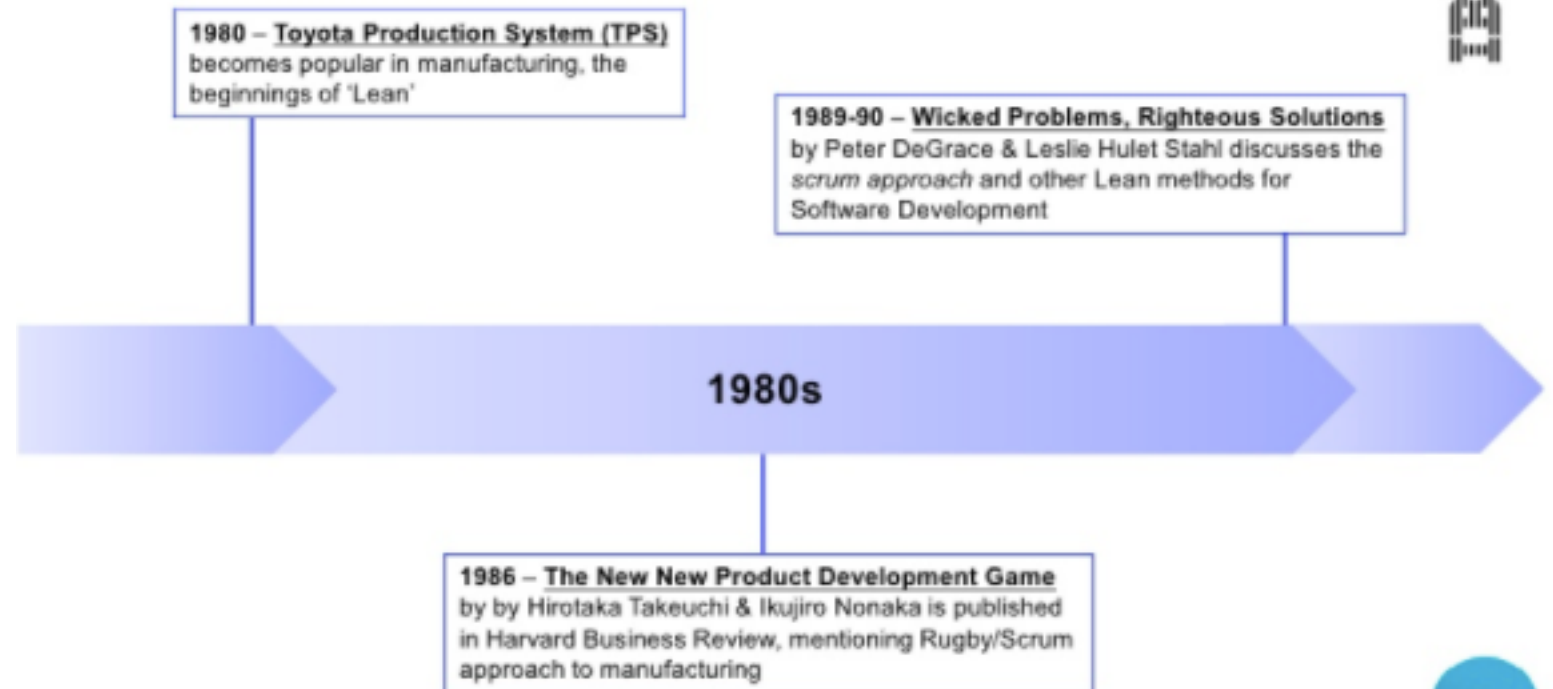
2010 – A Practical Guide to Distributed Scrum is published by: Elizabeth Woodward, Steffan Surdek, & Matthew Ganis

2001 – The Agile Manifesto is signed



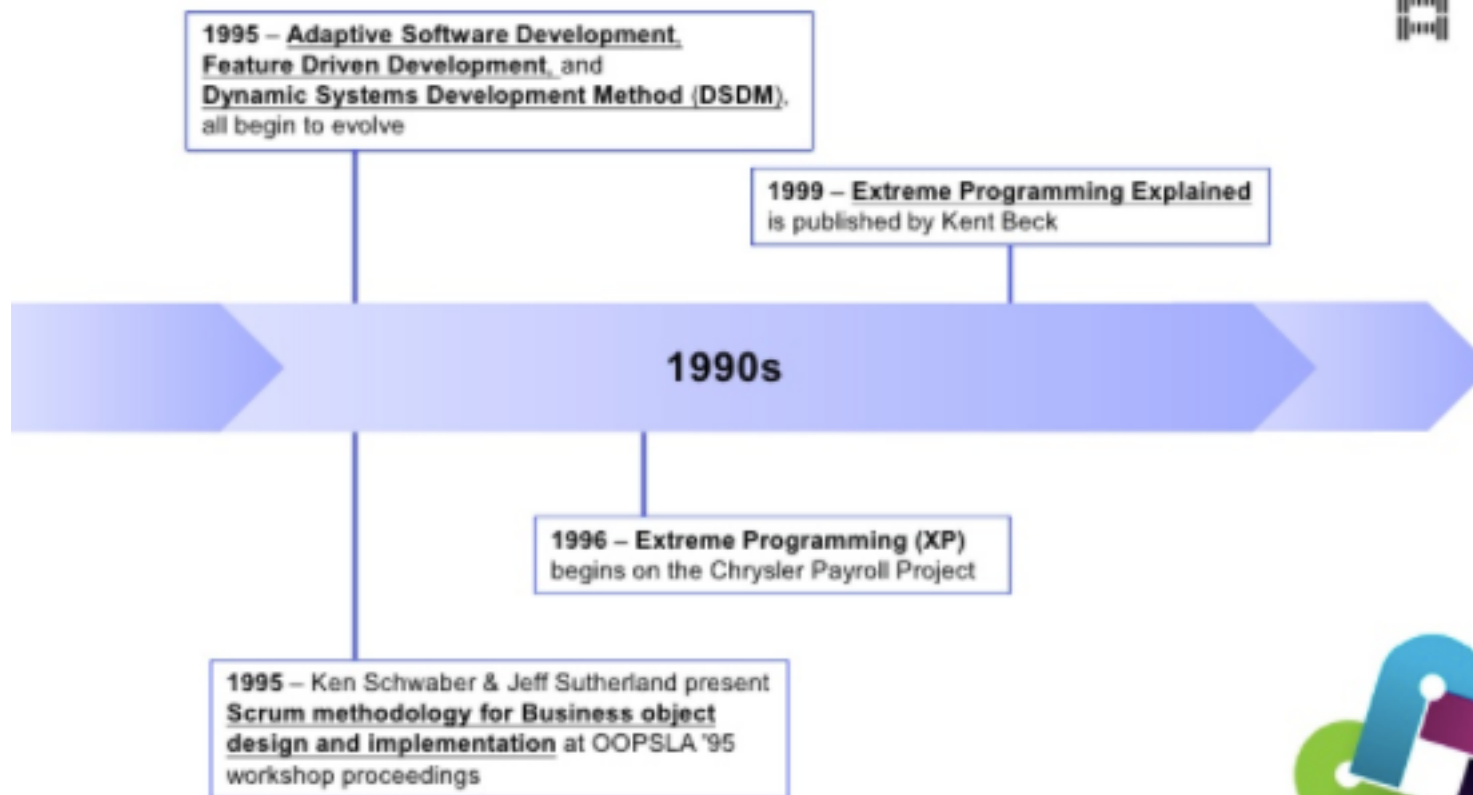
History of Agile

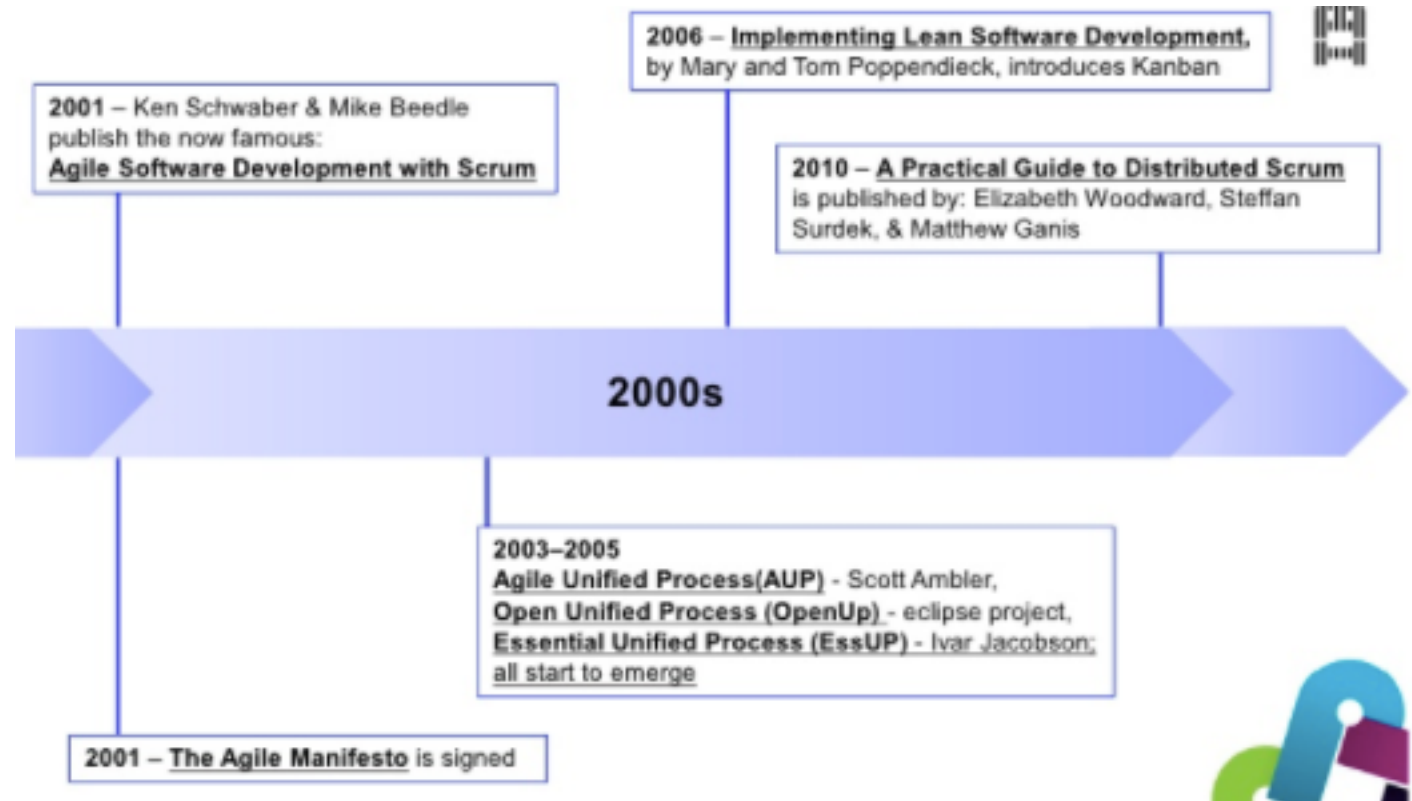
The important highlights as I see them...



History of Agile

The important highlights as I see them...

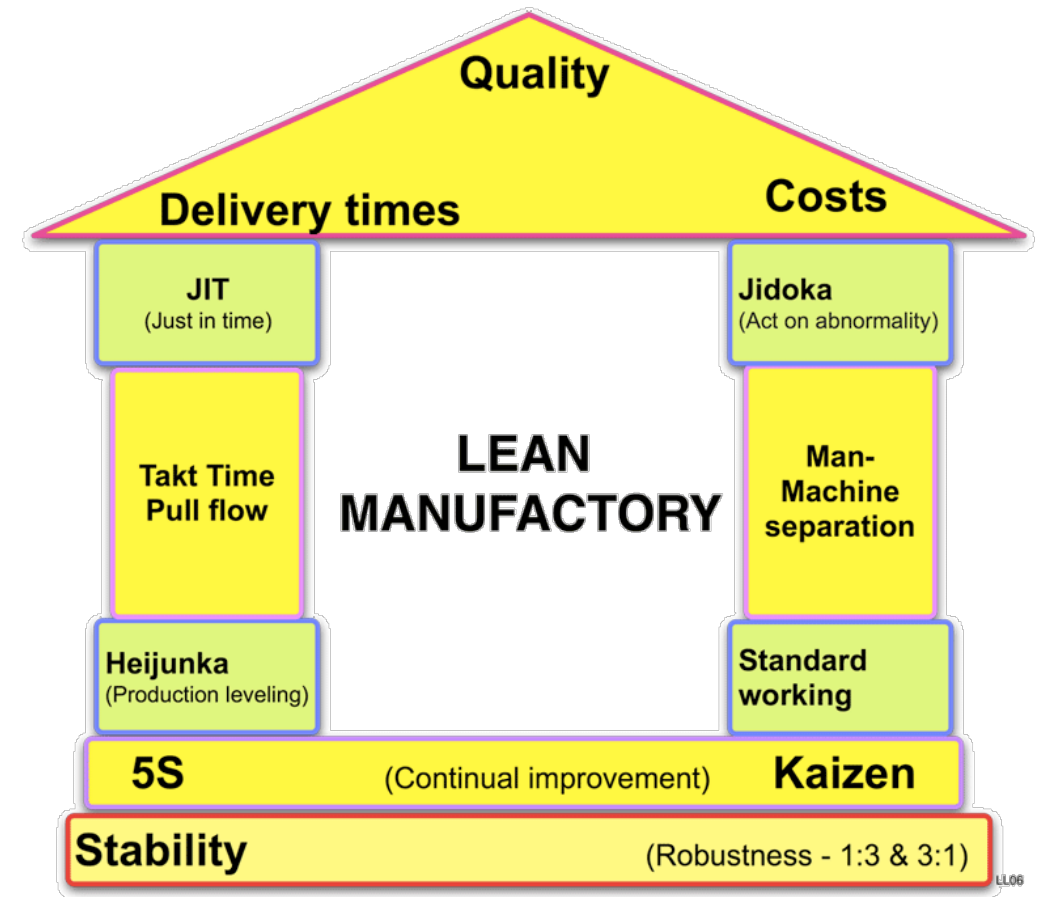




Lean Manufacturing

Lean manufacturing includes a set of principles that lean thinkers use to **achieve improvements** in productivity, quality, and lead-time **by eliminating waste through kaizen**. Kaizen is a Japanese word that essentially means "change for the better" or "good change."

<https://www.processexcellencenetwork.com/lean-six-sigma-business-transformation/articles/12-essential-lean-concepts-and-tools>



Agile Manifesto

The Agile Manifesto – a statement of values

Individuals and
interactions

over

Process and tools

Working software

over

Comprehensive
documentation

Customer
collaboration

over

Contract negotiation

Responding to
change

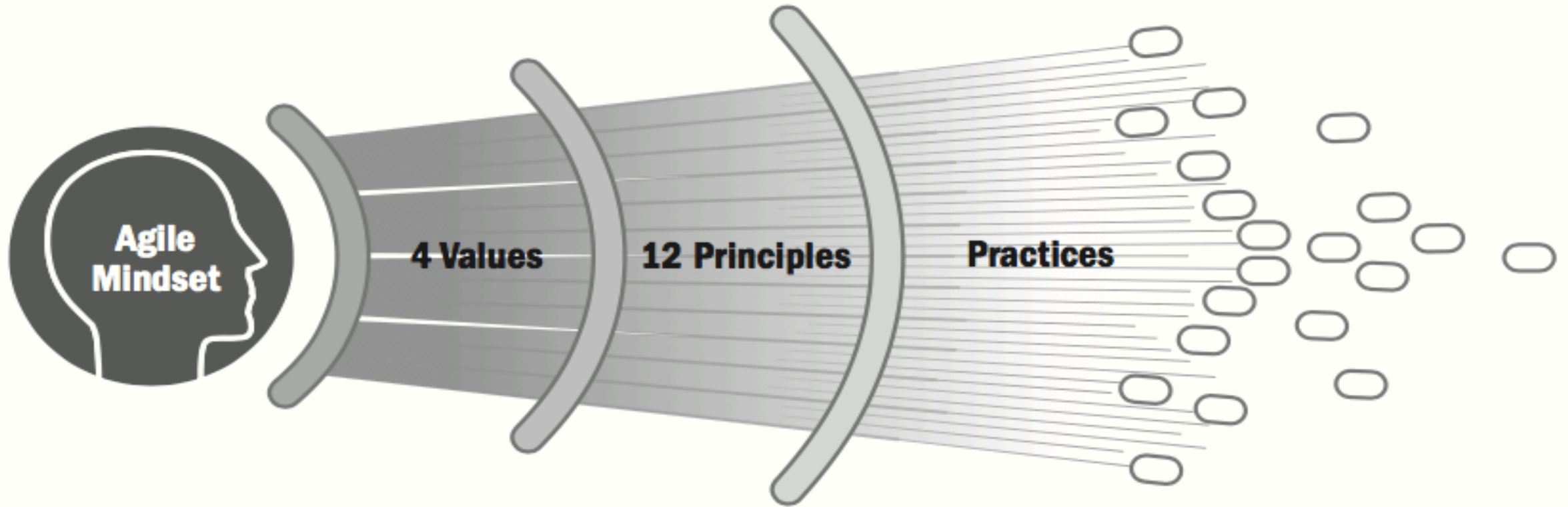
over

Following a plan

Agile Principles

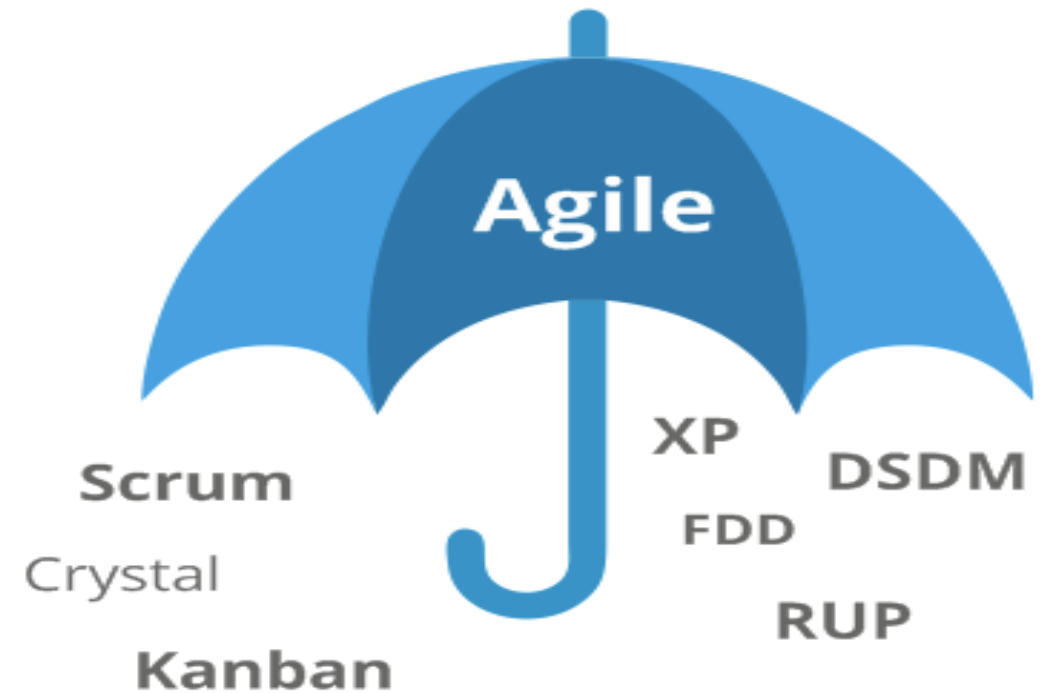
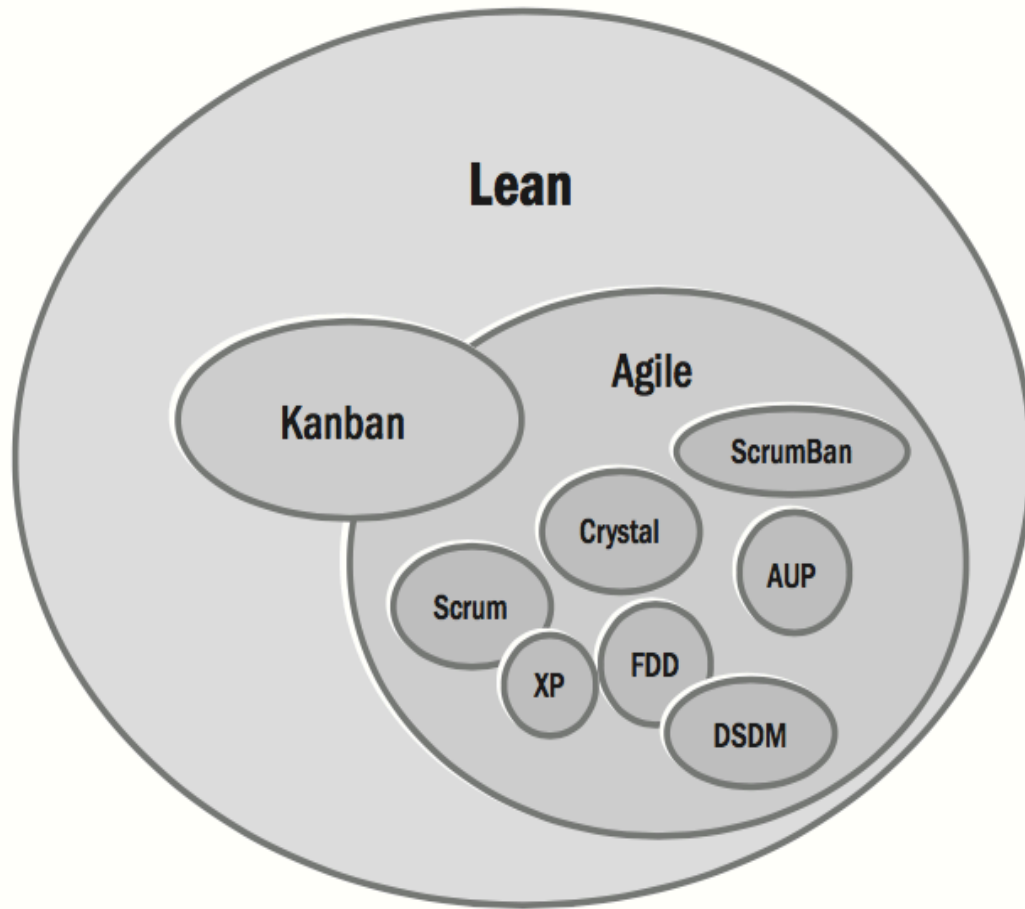
1. Our highest priority is to **satisfy the customer** through early and continuous delivery of valuable software.
2. **Welcome changing requirements**, even late in development. Agile processes harness change for the customer's competitive advantage.
3. **Deliver working software frequently**, from a couple of weeks to a couple of months, with a preference to the shorter timescale.
4. **Business people and developers must work together** daily throughout the project.
5. Build projects around **motivated individuals**.
Give them the environment and support they need, and trust them to get the job done.
6. The most efficient and effective method of conveying information to and within a development team is **face-to-face conversation**.
7. **Working software** is the primary measure of progress.
8. Agile processes **promote sustainable development**.
The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
9. **Continuous attention to technical excellence** and good design enhances agility.
10. Simplicity--the art of maximizing the amount of work not done--is essential.
11. **The best architectures**, requirements, and designs emerge from self-organizing teams.
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

Agile Mindset



Agile is a mindset defined by values, guided by principles, and manifested through many different practices. Agile practitioners select practices based on their needs.

Agile Umbrella



Scrum as an Agile Framework

Scrum is a framework for developing and sustaining complex products.

Source: scrumguides.org



Meet Jeff Sutherland

Jeff is the co-creator of Scrum and a leading expert on how the framework has evolved to meet the needs of today's business...

[Read More](#)



Meet Ken Schwaber

Ken Schwaber co-developed the Scrum process with Jeff Sutherland in the early 1990s to help organizations...

[Read More](#)

Use of Scrum

Initially developed for managing and developing products. Starting in the early 1990s, Scrum has been used extensively, worldwide, to:

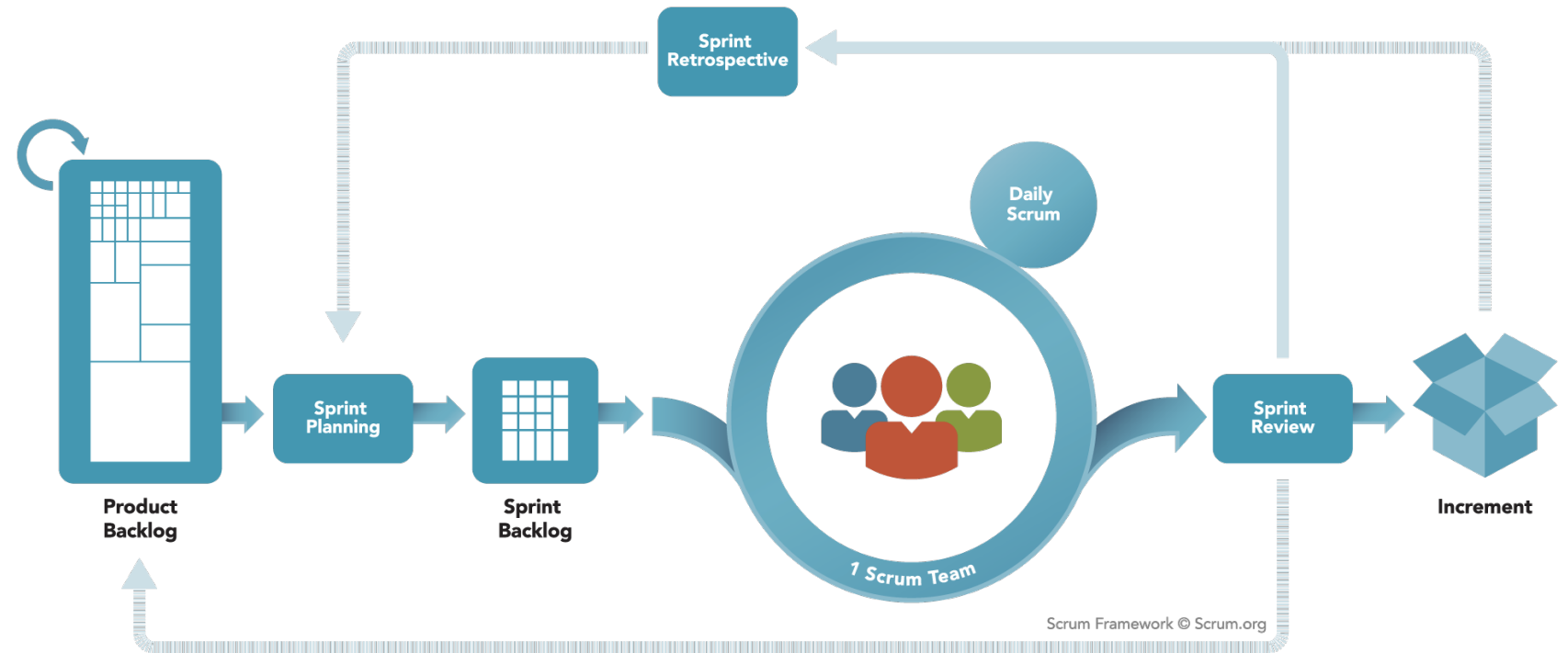
1. Research and identify viable markets, technologies, and product capabilities;
2. Develop products and enhancements;
3. Release products and enhancements, as frequently as many times per day;
4. Develop and sustain Cloud (online, secure, on-demand) and other operational environments for product use; and,
5. Sustain and renew products.

Scrum Team



Product Owner
Development Team
Scrum Master

Scrum Event



<https://www.eventbrite.hk/e/scrumorg-professional-scrum-master-course-jan-2018-tickets-37829125934>

<http://kshitij>

Product Backlog

Product
Backlog Item
#1

Product
Backlog Item
#2

..

Sprint Backlog

Product Backlog
Item #2

Product Backlog
Item #4

..

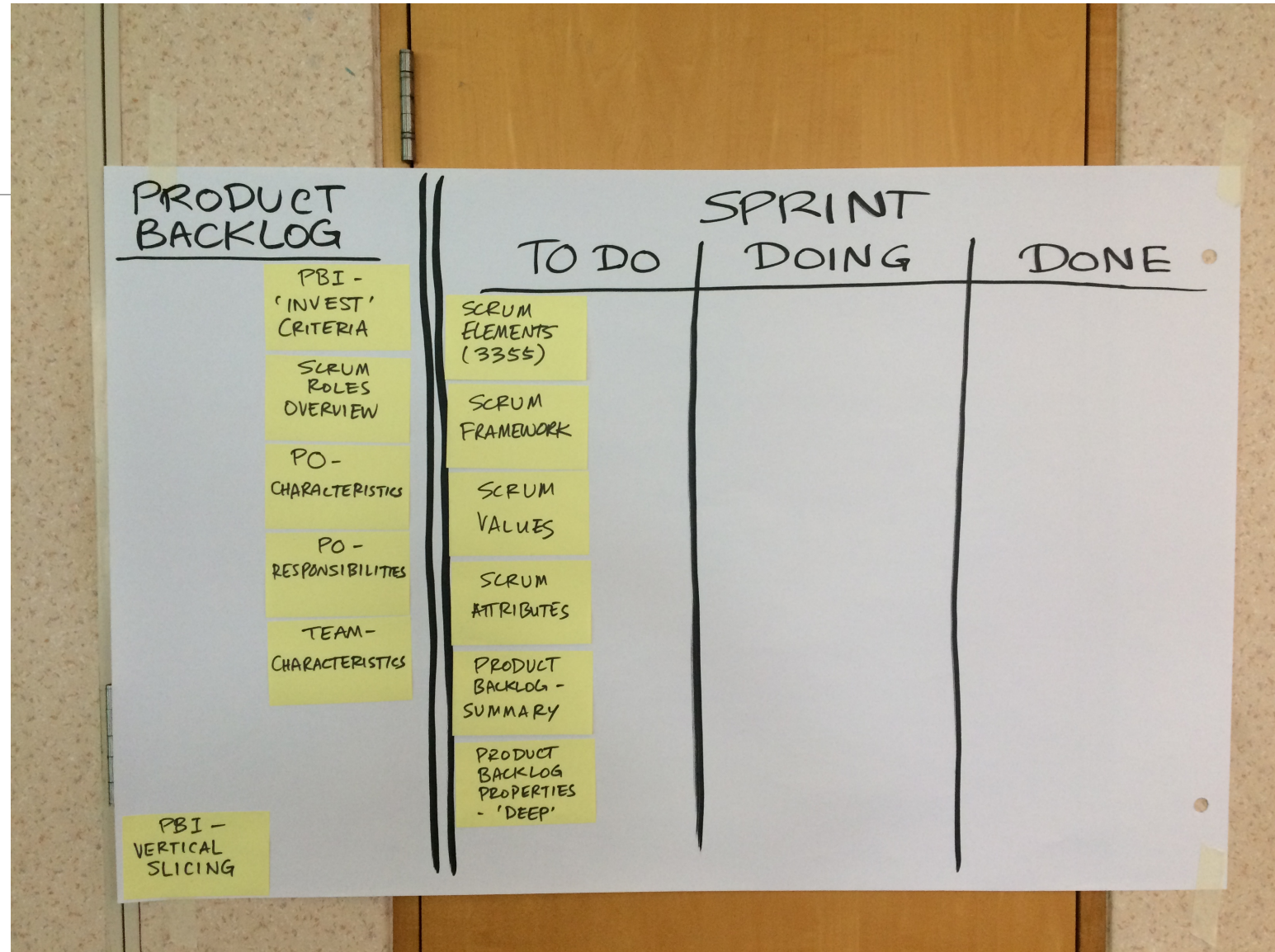
Tasks

Product Backlog
Item #1 - Task 1

Product Backlog
Item #1 - Taskt 2

...

Scrum Whiteboard



Scrum Tools



Kanban

Kanban is a scheduling system for lean manufacturing and just-in-time manufacturing (JIT). Kanban is an inventory control system for supply chains. Taiichi Ohno, an industrial engineer at Toyota, developed kanban to improve manufacturing efficiency. Kanban is one method to achieve JIT.

How Kanban Works

- 1. Visualize Work*
- 2. Limit Work in Process*
- 3. Focus on Flow*
- 4. Continuous Improvement*

SCRUM



Work is done within time-boxed sprints, generally of 2-4 weeks. The goal is to produce a potentially shippable product after each sprint.



Product is released on a particular cadence, which is determined by the sprints' length. So a team may release after 3 sprints, or every 6 weeks.



There's a heavy focus on cross-functionality. Teams have no specified roles; everyone is a "marketer."



Sprint kickoffs, daily standups, sprint reviews, and sprint retrospectives are vital rituals within the Scrum process.

KANBAN



There are no fixed-length sprints. Instead teams pull tasks from a prioritized backlog of things that need to be done.



Releases occur continuously, or whenever there is a shippable product created.



Team members can specialize and pull tasks related to their area of expertise, but too much specialization will reduce a team's effectiveness.



There is an emphasis on continually improving processes, but no standardized regular meetings or rituals.

Project Manager Roles

The role of the project manager in an agile project is somewhat of an **unknown**, because many agile frameworks and approaches do not address the role of the project manager

Project managers **shift from being the center to serving the team and the management**. In an agile environment, project managers are servant leaders, changing their emphasis to coaching people who want help, fostering greater collaboration on the team, and aligning stakeholder needs. As a servant leader, project managers encourage the distribution of responsibility to the team: to those people who have the knowledge to get work done.



Strategy to implement Agile

In general, there are **two strategies** to fulfill agile values and principles. **The first is to adopt a formal agile approach, intentionally designed and proven to achieve desired results.** Then take time to learn and understand the agile approaches before changing or tailoring them. Premature and haphazard tailoring can minimize the effects of the approach and thus limit benefits

The second strategy is to implement changes to project practices in a manner that fits the project context to achieve progress on a core value or principle. Use timeboxes to create features, or specific techniques to iteratively refine features. Consider dividing up one large project into several releases, if that works for the specific project context. Implement changes that will help the project succeed—the changes do not need to be part of the organization's formal practices. The end goal is not to be agile for its own sake, but rather to deliver a continuous flow of value to customers and achieve better business outcomes.



SESSION 4



TRADITIONAL OR AGILE APPROACH

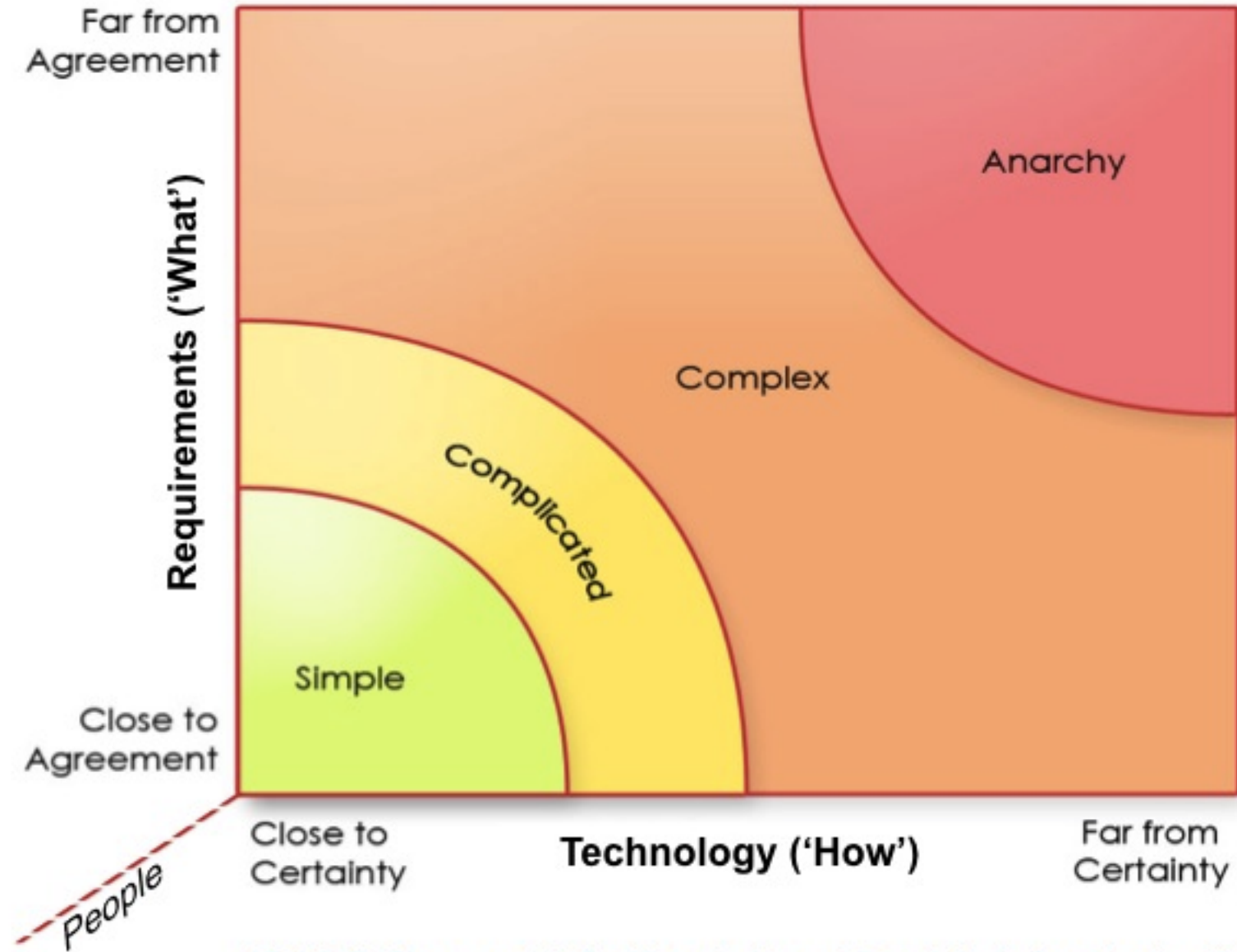
WHAT TO CALL NON-AGILE APPROACHES?

There is no single term that is universally used to describe non-agile approaches.

Some people prefer the terms *waterfall* or *serial* to describe this life cycle.

PROJECT UNCERTAINTY

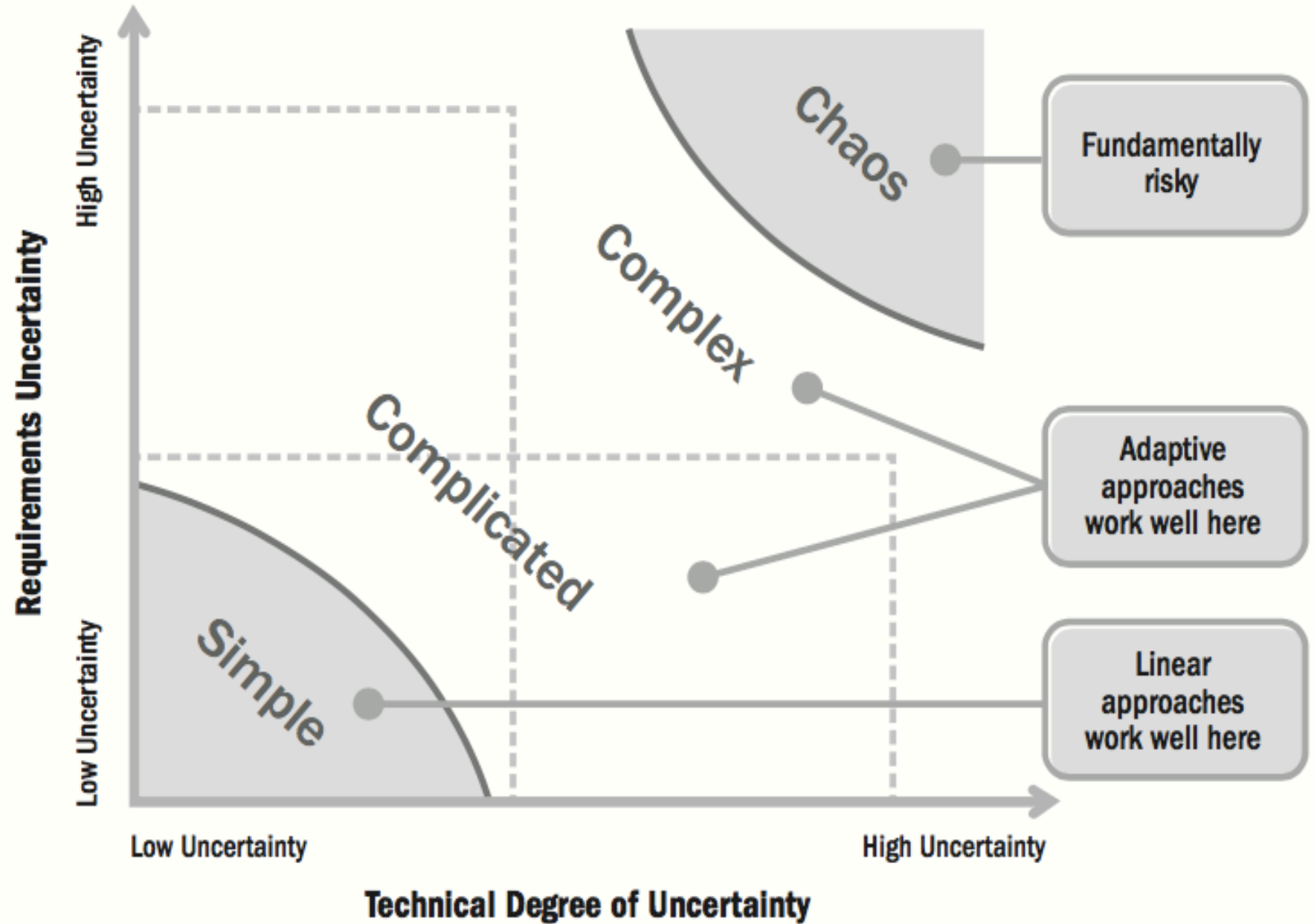
Some projects have considerable uncertainty around project requirements and how to fulfill those requirements using current knowledge and technology.



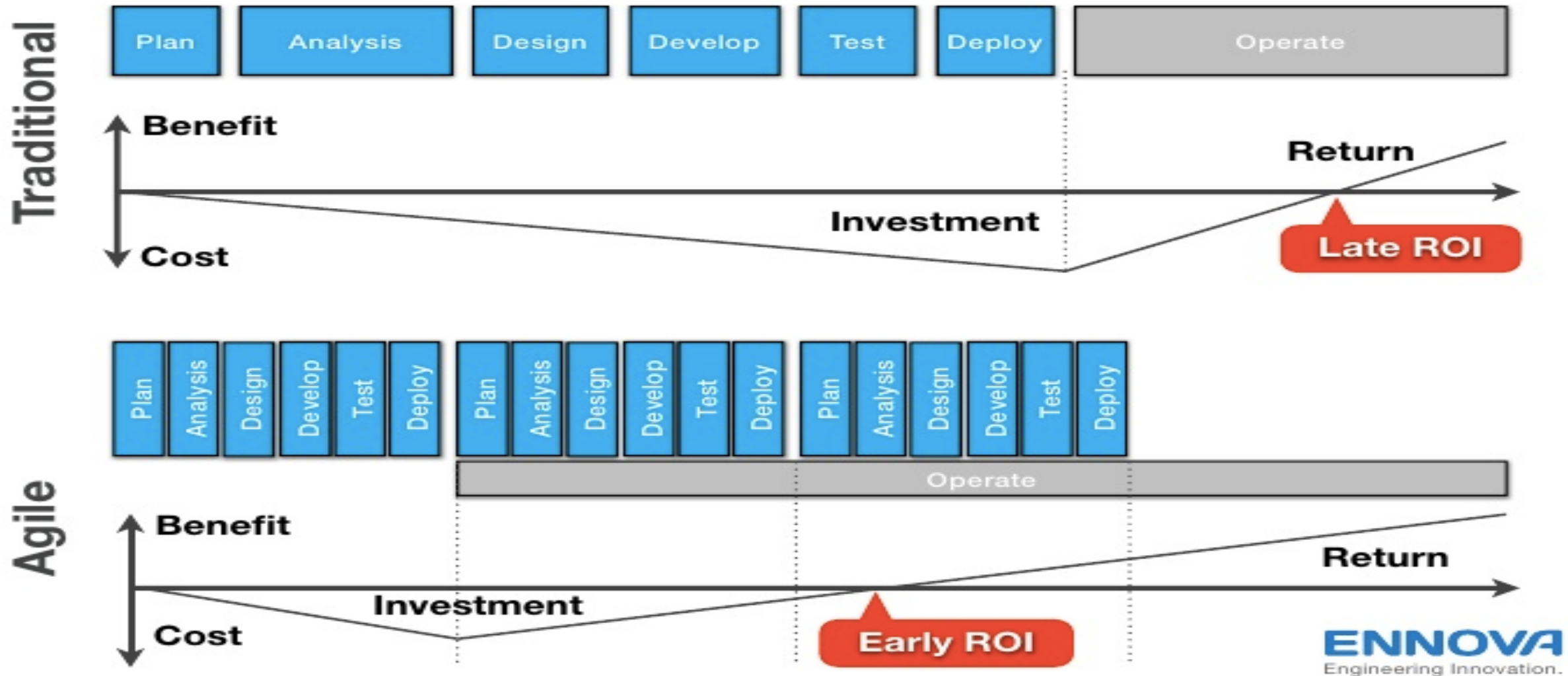
(Ralph D. Stacey matrix for "Complexity and Creativity in Organizations")



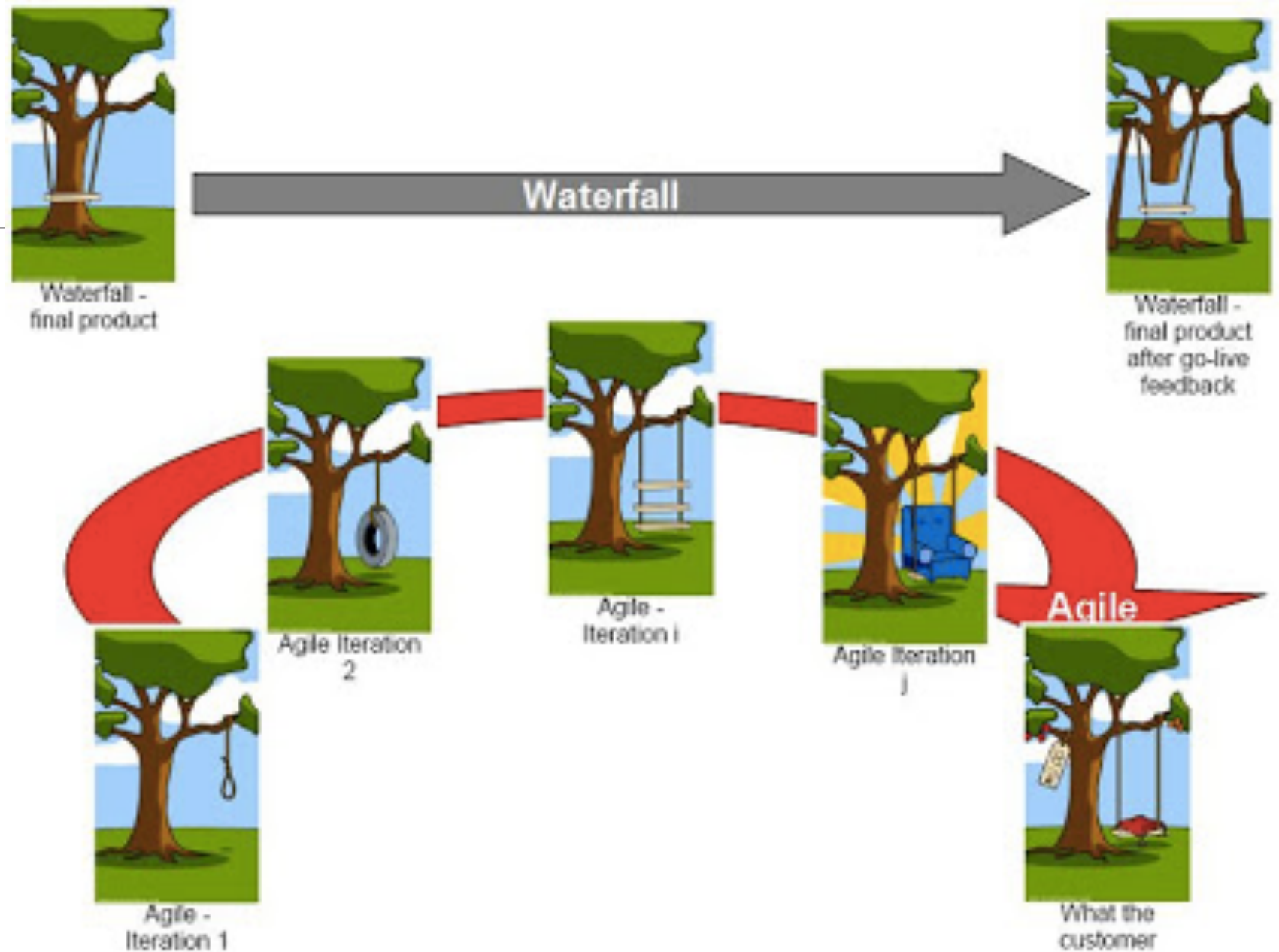
k18408071 fotosearch.com ©



Traditional vs Agile



Traditional vs Agile



Constraints

**Predictive Process
(Waterfall)**

**Adaptive Process
(Agile)**

Requirements

Cost

Schedule

**Plan
Driven**

**Value/
Vision
Driven**

Estimates

Cost

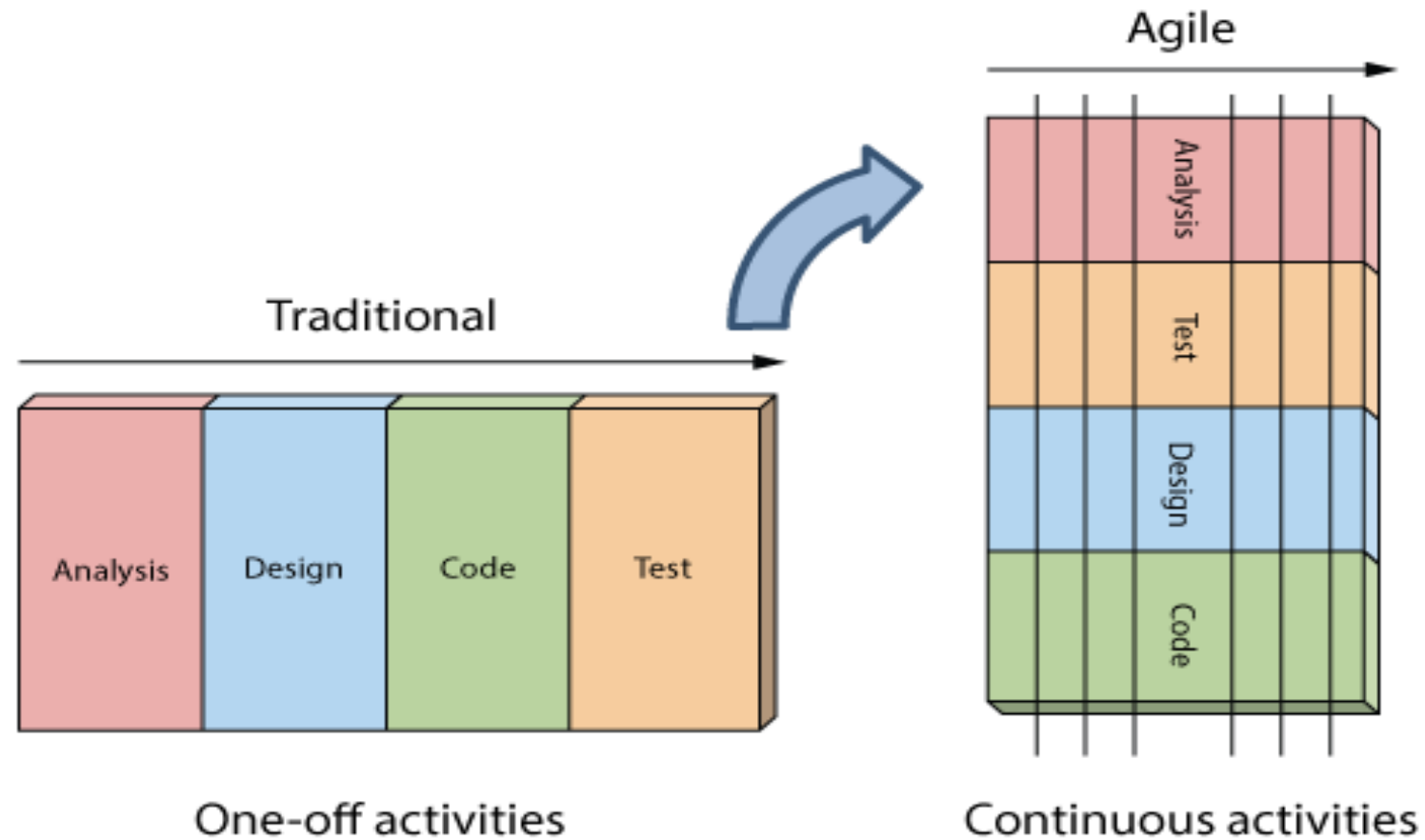
Schedule

Features

The plan creates
cost/schedule estimates

The vision creates
feature estimates

Traditional vs Agile



Advantage of Waterfall Approach



1. Developers and customers agree on what will be delivered early in the development lifecycle. This makes **planning and designing more straightforward**.
2. **Progress is more easily measured**, as the full scope of the work is known in advance.
3. Throughout the development effort, **it's possible for various members of the team to be involved or to continue with other work, depending on the active phase of the project**.
4. Except for reviews, approvals, status meetings, etc., a customer presence is not strictly required after the requirements phase.
5. Finally, **the software can be designed completely and more carefully**, based upon a more complete understanding of **all** software deliverables. This provides a better software design with less likelihood of the “piecemeal effect,” a development phenomenon that can occur as pieces of code are defined and subsequently added to an application where they may or may not fit well.



Disadvantage of Agile Approach

The very high degree of customer involvement, while great for the project, may present problems for some customers who simply **may not have the time or interest for this type of participation.**

Agile works best when members of the development team **are completely dedicated to the project.**

Because Agile focuses on time-boxed delivery and frequent reprioritization, it's possible that some items set for delivery will not be completed within the allotted timeframe. Additional sprints (beyond those initially planned) may be needed, adding to the project cost.

The close working relationships in an Agile project are easiest to manage when the team members are **located in the same physical space**, which is not always possible. However, there are a variety of ways to handle this issue, such as webcams, collaboration tools, etc.

The iterative nature of Agile development **may lead to a frequent refactoring if the full scope of the system is not considered in the initial architecture and design.** Without this refactoring, the system can suffer from a reduction in overall quality. This becomes more pronounced in larger-scale implementations, or with systems that include a high level of integration.

LIFE CYCLE SELECTION

- 1. Predictive life cycle.** A more traditional approach, with the bulk of planning occurring upfront, then executing in a single pass; a sequential process.
- 2. Iterative life cycle.** An approach that allows feedback for unfinished work to improve and modify that work.
- 3. Incremental life cycle.** An approach that provides finished deliverables that the customer may be able to use immediately.
- 4. Agile life cycle.** An approach that is both iterative and incremental to refine work items and deliver frequently.

Traditional vs Agile

Characteristics				
Approach	Requirements	Activities	Delivery	Goal
Predictive	Fixed	Performed once for the entire project	Single delivery	Manage cost
Iterative	Dynamic	Repeated until correct	Single delivery	Correctness of solution
Incremental	Dynamic	Performed once for a given increment	Frequent smaller deliveries	Speed
Agile	Dynamic	Repeated until correct	Frequent small deliveries	Customer value via frequent deliveries and feedback

Predictive Life Cycle

Predictive life cycles expect to take advantage of high certainty around firm requirements, a stable team, and low risk. As a result, project activities often execute in a serial manner,

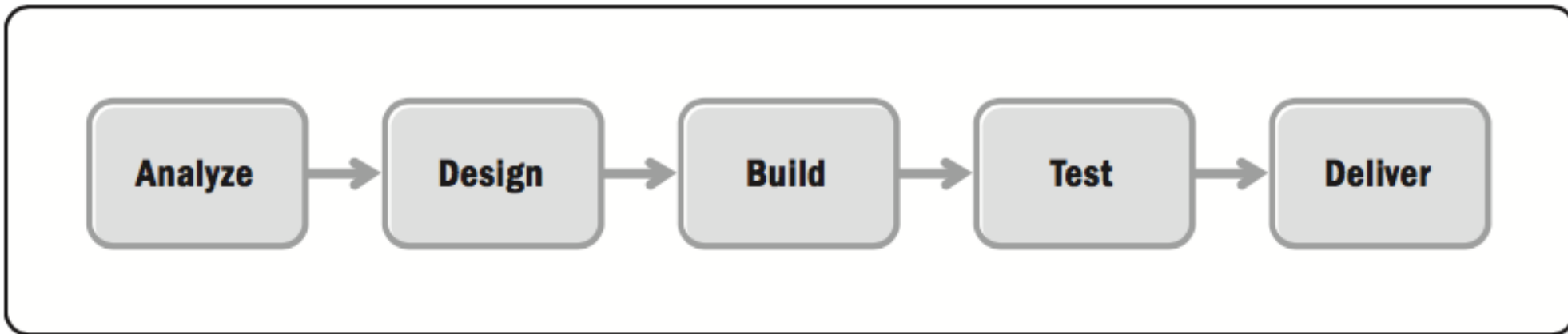


Figure 3-2. Predictive Life Cycle

Iterative Life Cycle

Iterative life cycles improve the product or result through **successive prototypes or proofs of concept**. Each new prototype yields new stakeholder feedback and team insights. Then, the team incorporates the new information by repeating one or more project activities in the next cycle

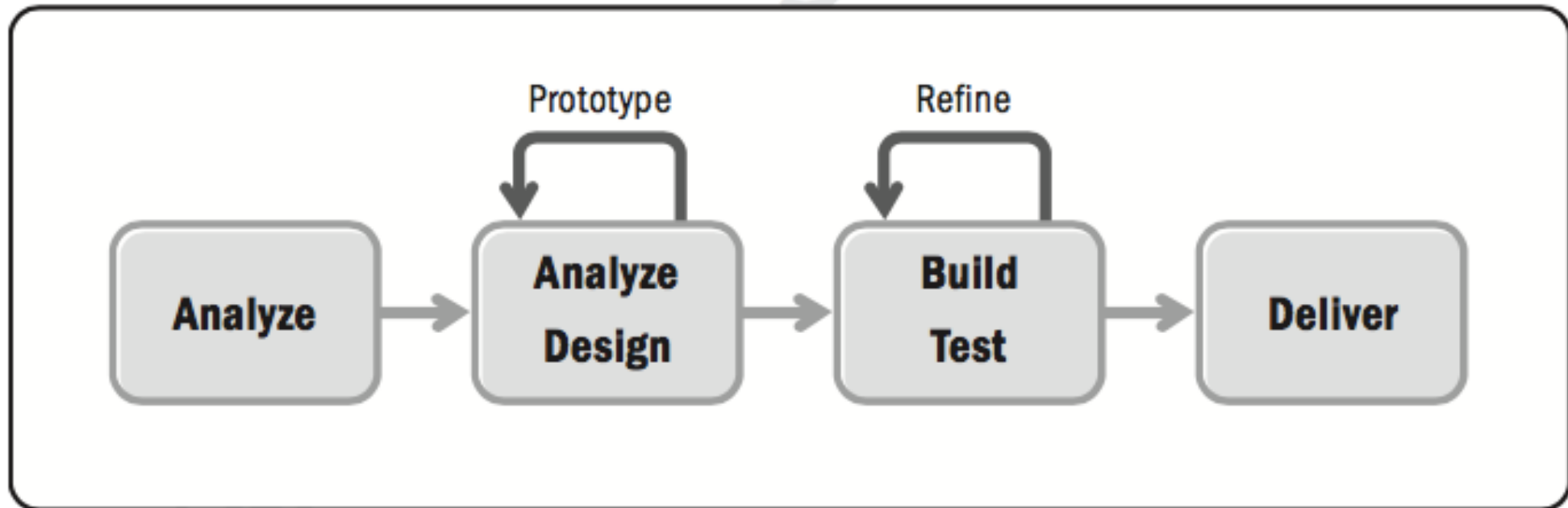


Figure 3-3. Iterative Life Cycle

Incremental Life Cycle

Some projects optimize for speed of delivery. Many businesses and initiatives cannot afford to wait for everything to be completed; in these cases, customers are willing to receive **a subset of the overall solution**. This frequent delivery of smaller deliverables is called an incremental life cycle

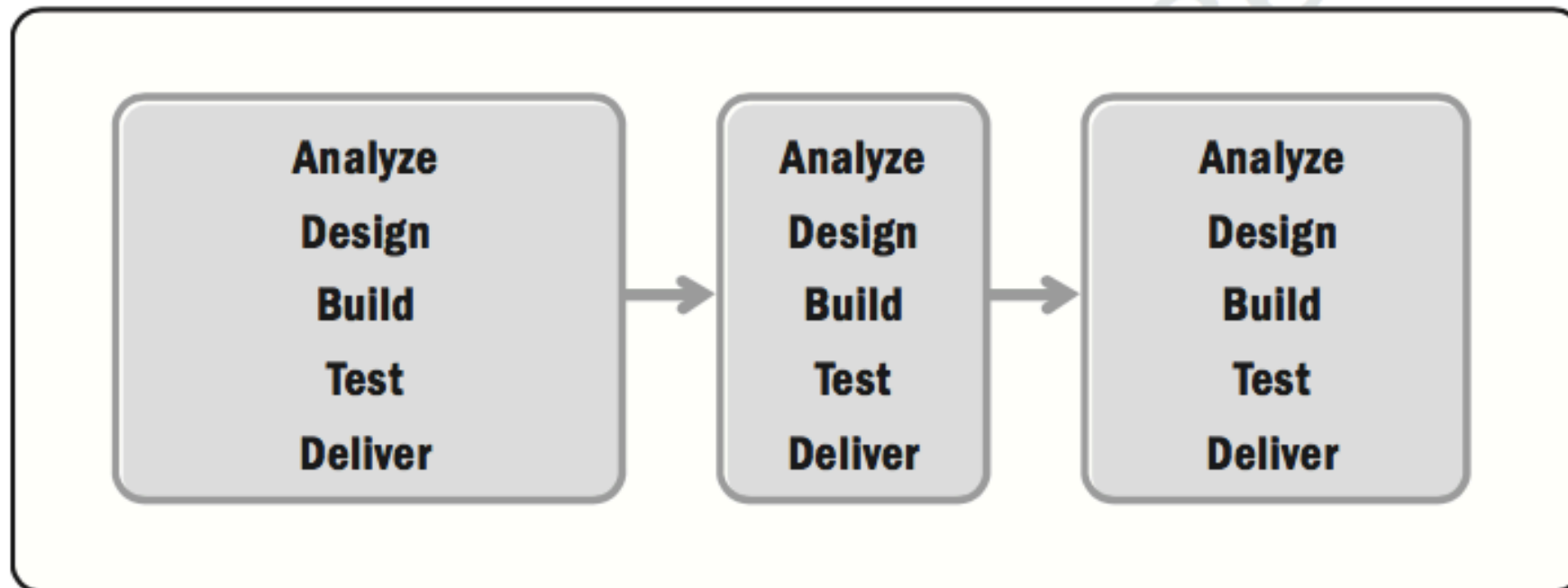
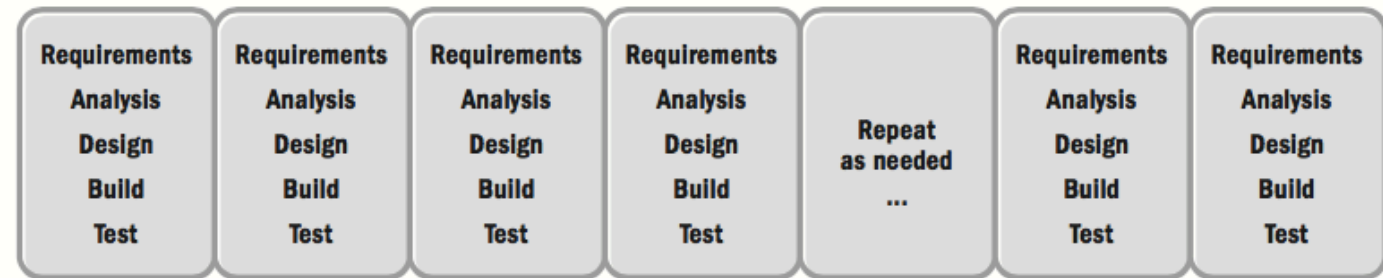


Figure 3-4. A Life Cycle of Varying-Sized Increments

Agile Life Cycle

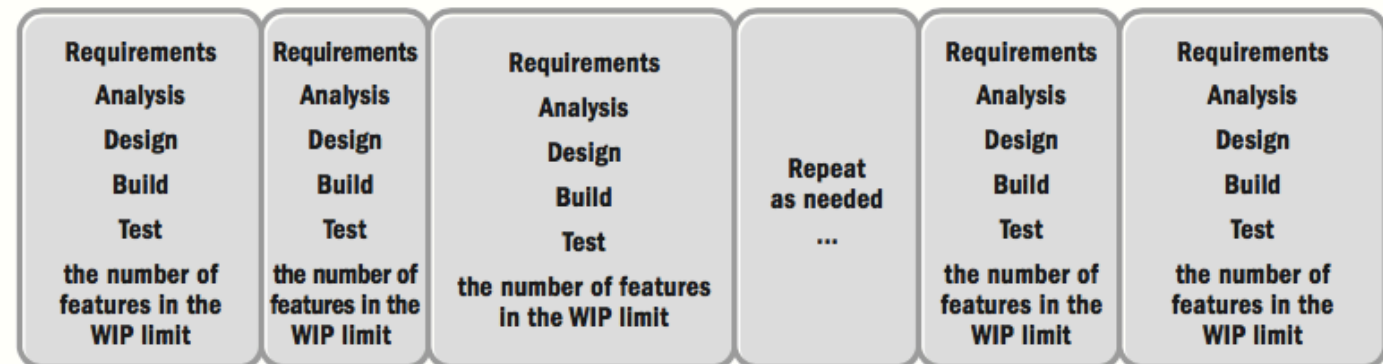
In an agile environment, **the team expects requirements to change**. The iterative and incremental approaches provide feedback to better plan the next part of the project. However, in agile projects, incremental delivery uncovers hidden or misunderstood requirements.

Iteration-Based Agile



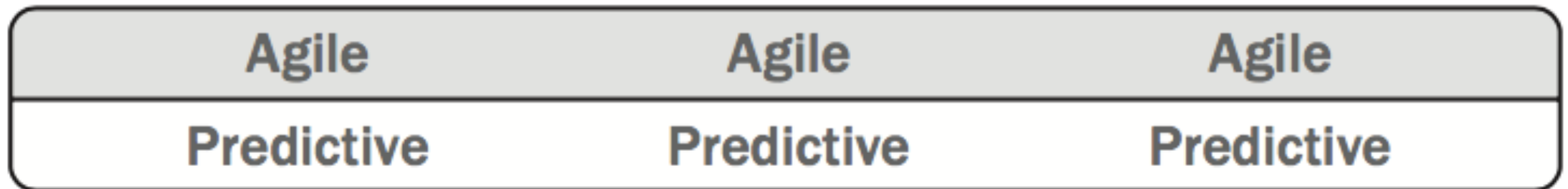
NOTE: Each timebox is the same size. Each timebox results in working tested features.

Flow-Based Agile

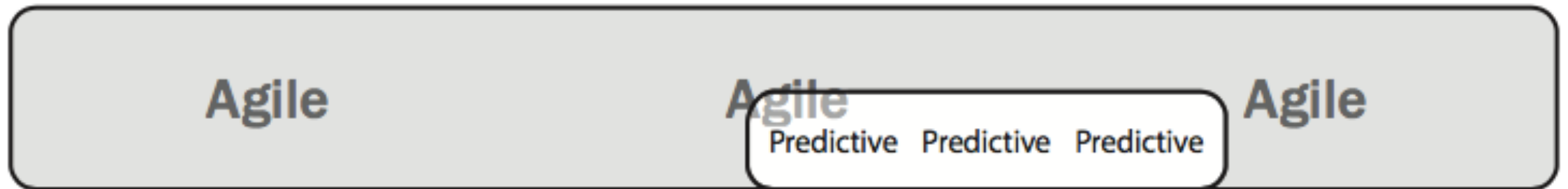
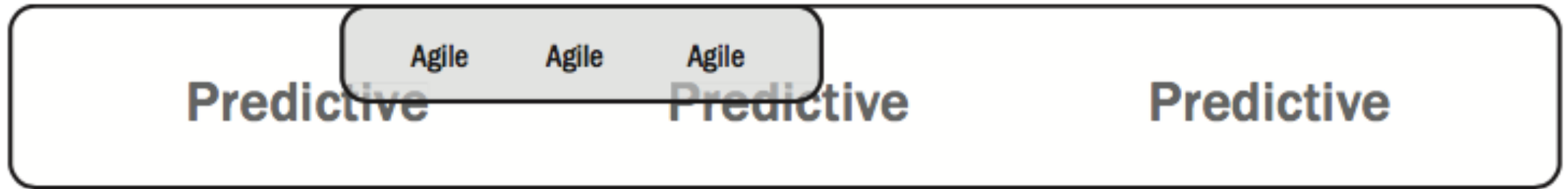


NOTE: In flow, the time it takes to complete a feature is not the same for each feature.

Hybrid Life Cycle



Hybrid Life Cycle



Traditional or Agile ?

1. Construction Project
2. Data Centre Building
3. Hardware Delivery / Deployment
4. OS Software development
5. OS Installation
6. Database Installation
7. Upgrade patch software

1. Software / Product Development
 2. Startup Business
-
1. ERP Development
 2. ERP Customization
 3. Package Product Customization

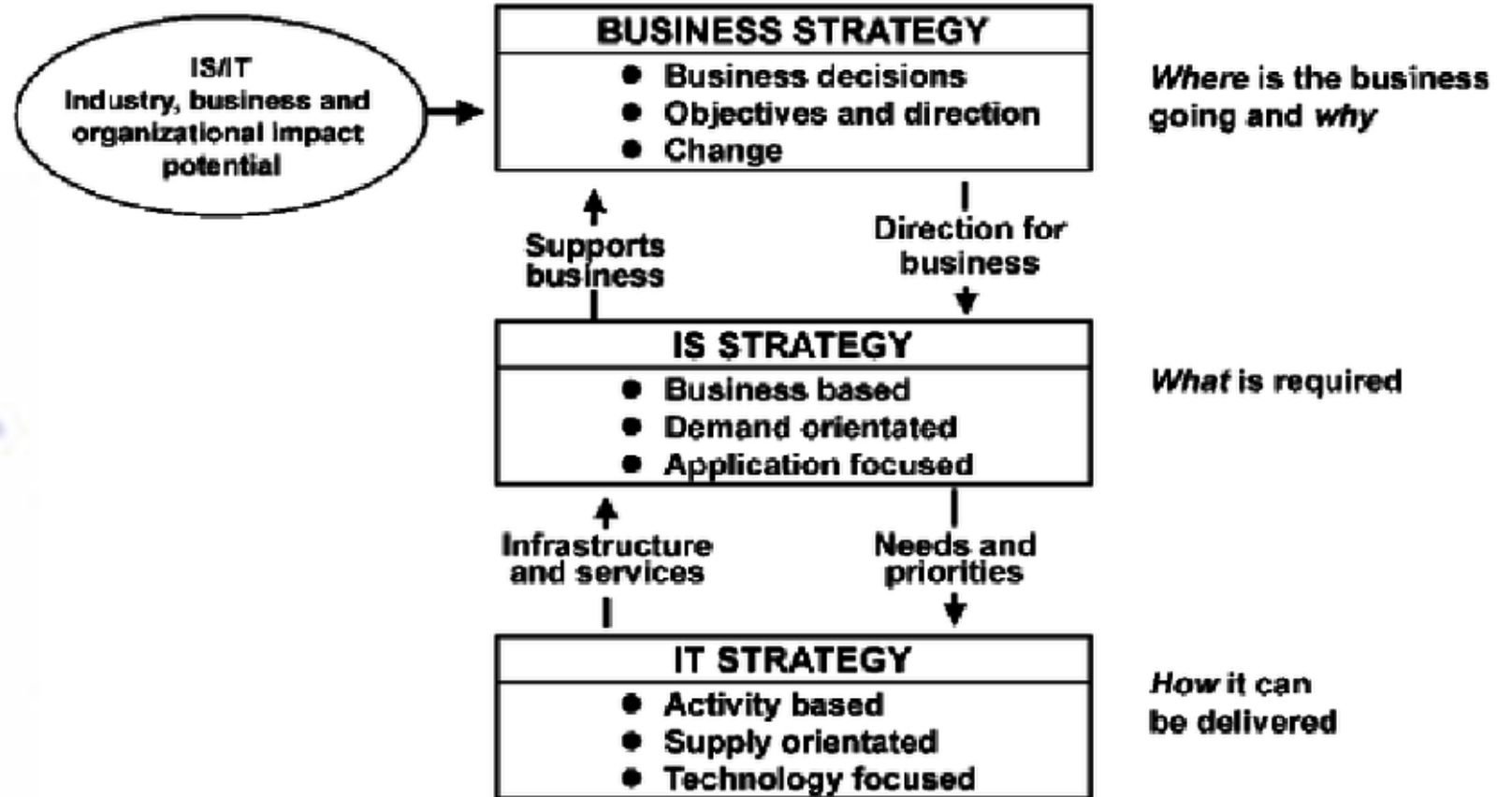
Which One is Better, Traditional or Agile ?

What is the characteristic of your project in your organization ?

Characteristics				
Approach	Requirements	Activities	Delivery	Goal
Predictive	Fixed	Performed once for the entire project	Single delivery	Manage cost
Iterative	Dynamic	Repeated until correct	Single delivery	Correctness of solution
Incremental	Dynamic	Performed once for a given increment	Frequent smaller deliveries	Speed
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Which One is Better, Traditional or Agile ?

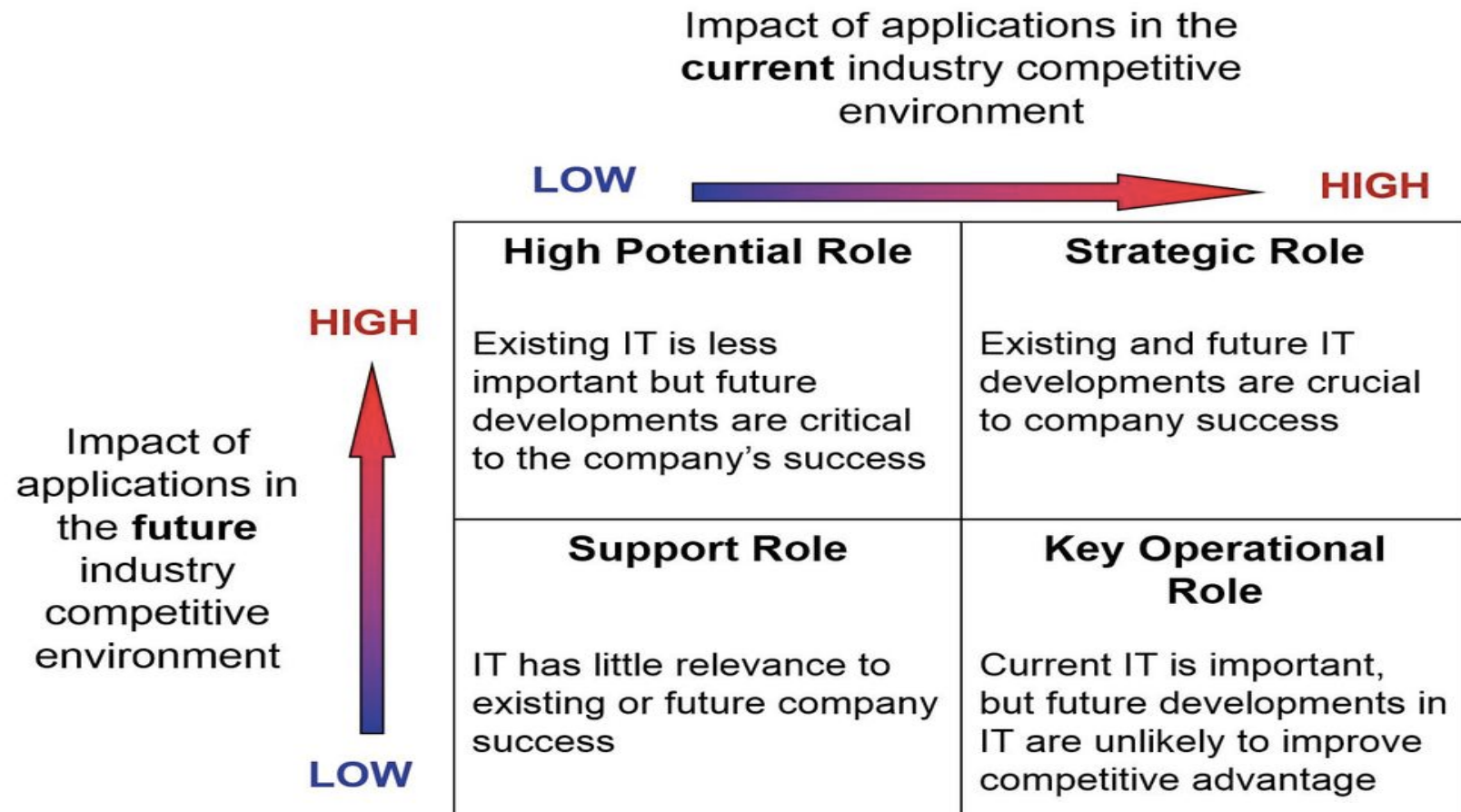
Aligning IS/IT Strategy with Organization Strategy



Which One is Better, Traditional or Agile ?

Application / Project Mapping

McFarlan's Grid



Options

